第8章 正弦稳态电路的分析

8.1 阻抗和导纳

8.2 正弦稳态电路的分析

8.3 正弦稳态电路的功率

8.4 复功率

8.5 最大功率传输

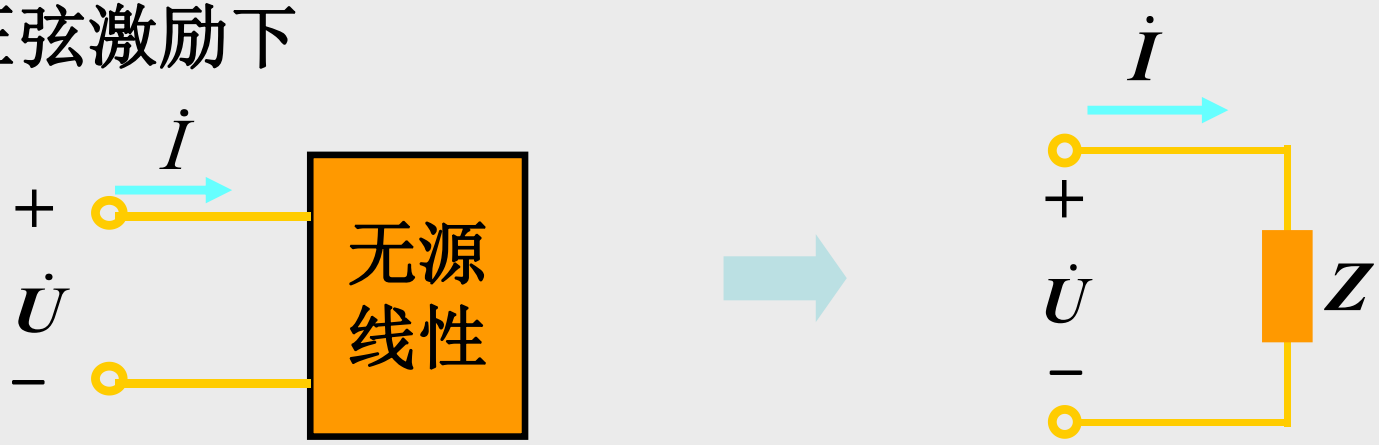
重点：

1. 阻抗和导纳；
2. 正弦稳态电路的分析；
3. 正弦稳态电路的功率分析。

8.1 阻抗和导纳

1. 阻抗

正弦激励下



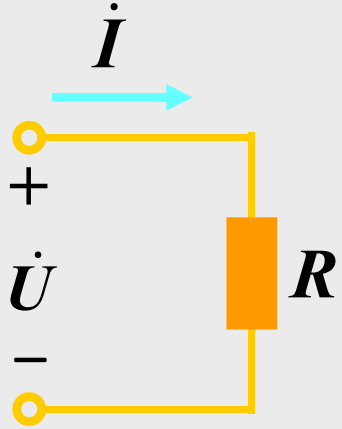
定义阻抗 $Z=\dfrac{\dot{U}}{\dot{I}}=|Z|\angle\varphi$

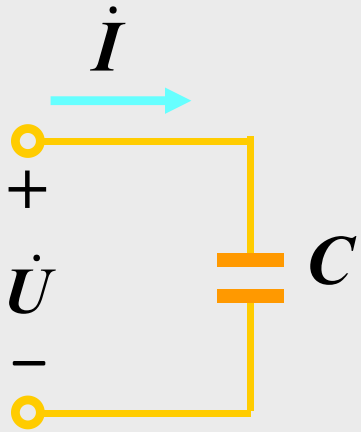
欧姆定律的相量形式

$|Z|=\dfrac{U}{I}$　　阻抗模　　单位：Ω

$\varphi=\psi_u-\psi_i$　　阻抗角

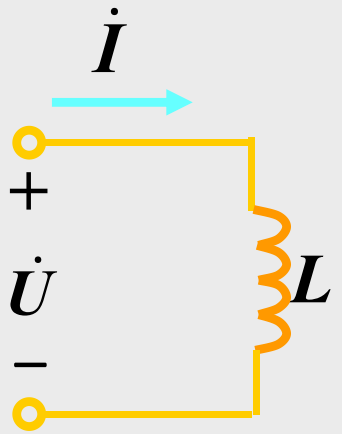
当无源网络内为单个元件时有：





$$Z=\frac{\dot{U}}{\dot{I}}=R$$

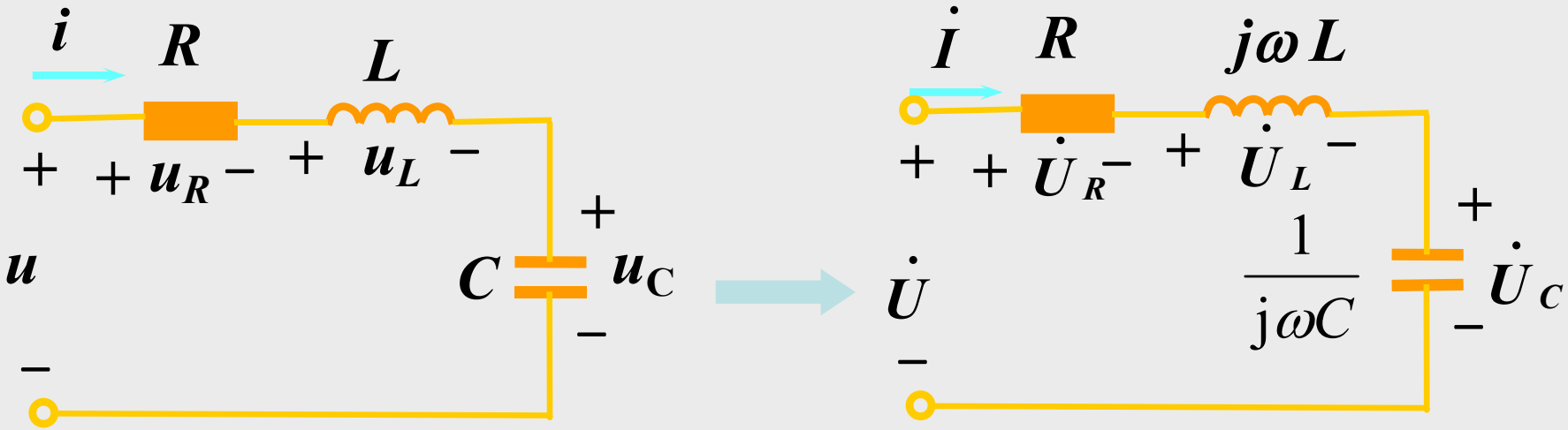
$$Z=\frac{\dot{U}}{\dot{I}}=-j\frac{1}{\omega C}=jX_C \qquad X_C=-\frac{1}{\omega C}$$



$$Z=\frac{\dot{U}}{\dot{I}}=j\omega L=jX_L \qquad X_L=\omega L$$

Z可以是实数，也可以是虚数

2. *RLC*串联电路



由KVL：
$$\dot{U}=\dot{U}_R+\dot{U}_L+\dot{U}_C=R\dot{I}+\mathrm{j}\omega L\dot{I}-\mathrm{j}\frac{1}{\omega C}\dot{I}$$

$$=[R+j(\omega L-\frac{1}{\omega C})]\dot{I}=[R+j(X_L+X_C)]\dot{I}=(R+jX)\dot{I}$$

$$Z=\frac{\dot{U}}{\dot{I}}=R+j\omega L-j\frac{1}{\omega C}=R+jX=|Z|\angle\varphi$$

Z— 复阻抗；R—电阻(阻抗的实部)；X—电抗(阻抗的虚部)；

$|Z|$—复阻抗的模；φ—阻抗角。

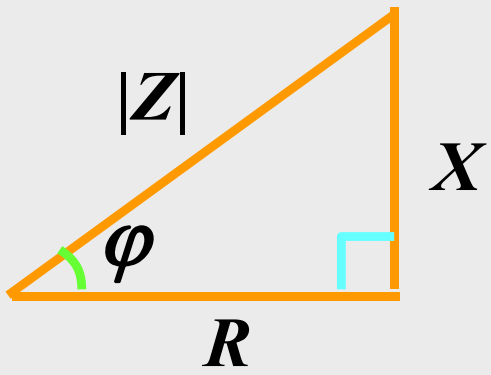
关系：

$$\begin{cases} |Z| = \sqrt{R^2 + X^2} \\ \varphi = \text{arctg}\dfrac{X}{R} \end{cases}$$

或

$$\begin{cases} R = |Z|\cos\varphi \\ X = |Z|\sin\varphi \end{cases} \qquad |Z| = \frac{U}{I} \qquad \varphi = \psi_u - \psi_i$$

阻抗三角形



分析 R、L、C 串联电路得出：

（1）$Z=R+j(\omega L-1/\omega C)=|Z|\angle\varphi$ 为复数，故称复阻抗

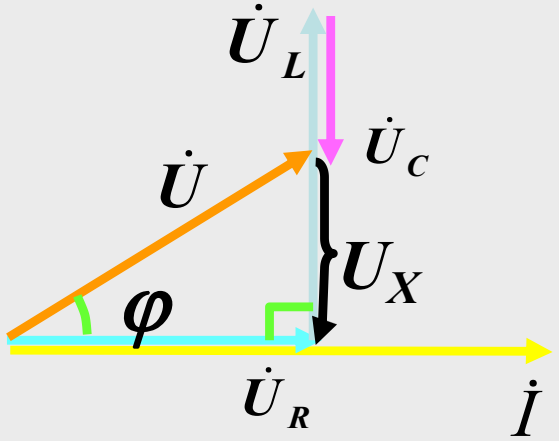
（2）$\omega L > 1/\omega C$，$X>0$，$\varphi>0$，电路为感性，电压领先电流；

$\omega L<1/\omega C$，$X<0$，$\varphi<0$，电路为容性，电压落后电流；

$\omega L=1/\omega C$，$X=0$，$\varphi=0$，电路为电阻性，电压与电流同相。

（3）相量图：选电流为参考向量，$\psi_i = 0$

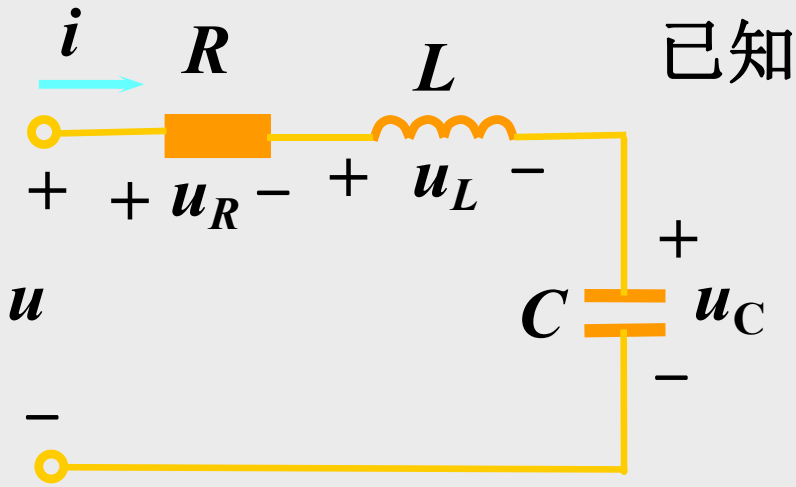
设$\omega L > 1/\omega C$



三角形U_R、U_X、U 称为电压三角形，它和阻抗三角形相似。即

$$U=\sqrt{U_R^2+U_X^2}$$

例



已知：R=15Ω, L=0.3mH, C=0.2μF,

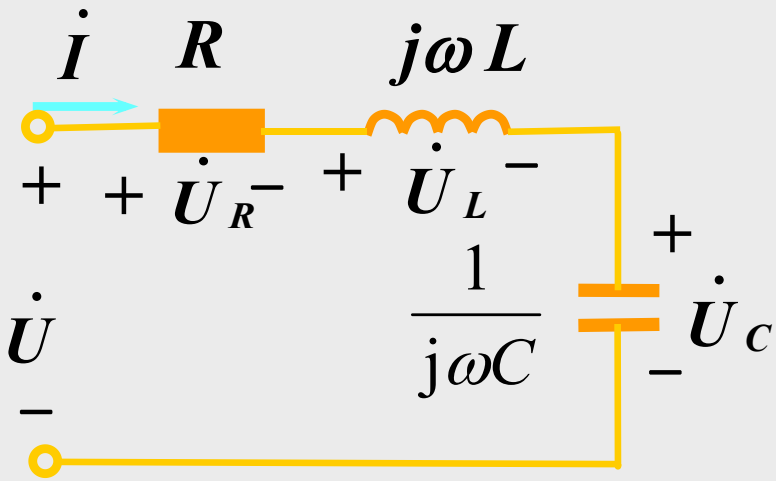
$$u = 5\sqrt{2}\cos(\omega t + 60^\circ)$$

$$f = 3\times10^4\,\text{Hz}\,.$$

求 i, u_R, u_L, u_C.

解

其相量模型为：



$$\dot{U} = 5\angle 60^\circ\ \text{V}$$

$$\text{j}\omega L = \text{j}2\pi\times3\times10^4\times0.3\times10^{-3} = \text{j}56.5\Omega$$

$$-\text{j}\frac{1}{\omega C} = -\text{j}\frac{1}{2\pi\times3\times10^4\times0.2\times10^{-6}} = -\text{j}26.5\Omega = 26.5\angle-90^\circ\Omega$$

$$Z = R + \text{j}\omega L - \text{j}\frac{1}{\omega C} = 15 + \text{j}56.5 - \text{j}26.5 = 33.54\angle63.4^\circ\,\Omega$$

$$\dot{I}=\frac{\dot{U}}{Z}=\frac{5\angle 60^\circ}{33.54\angle 63.4^\circ}=0.149\angle -3.4^\circ \text{ A}$$

$$\dot{U}_R=R\dot{I}=15\times 0.149\angle -3.4^\circ=2.235\angle -3.4^\circ \text{ V}$$

$$\dot{U}_L=\mathrm{j}\omega L\dot{I}=56.5\angle 90^\circ\times 0.149\angle -3.4^\circ=8.42\angle 86.6^\circ \text{ V}$$

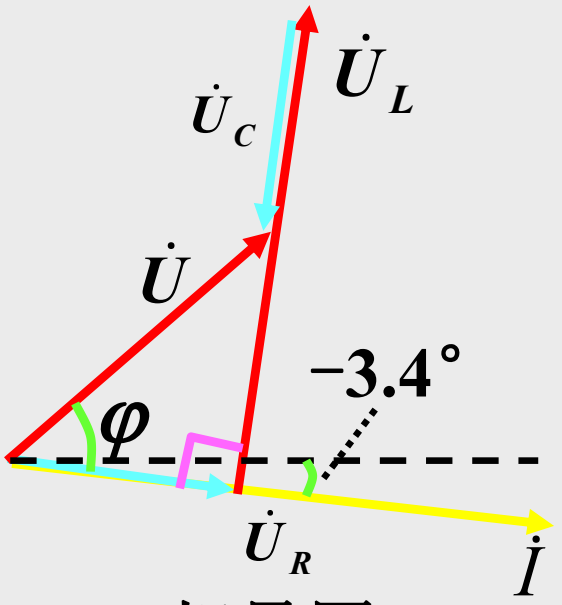
$$\dot{U}_C=\mathrm{j}\frac{-1}{\omega C}\dot{I}=26.5\angle -90^\circ\times 0.149\angle -3.4^\circ=3.95\angle -93.4^\circ \text{ V}$$

则 $i=0.149\sqrt{2}\cos(\omega t-3.4^\circ)$ A

$u_R=2.235\sqrt{2}\cos(\omega t-3.4^\circ)$ V

$u_L=8.42\sqrt{2}\cos(\omega t+86.6^\circ)$ V

$u_C=3.95\sqrt{2}\cos(\omega t-93.4^\circ)$ V

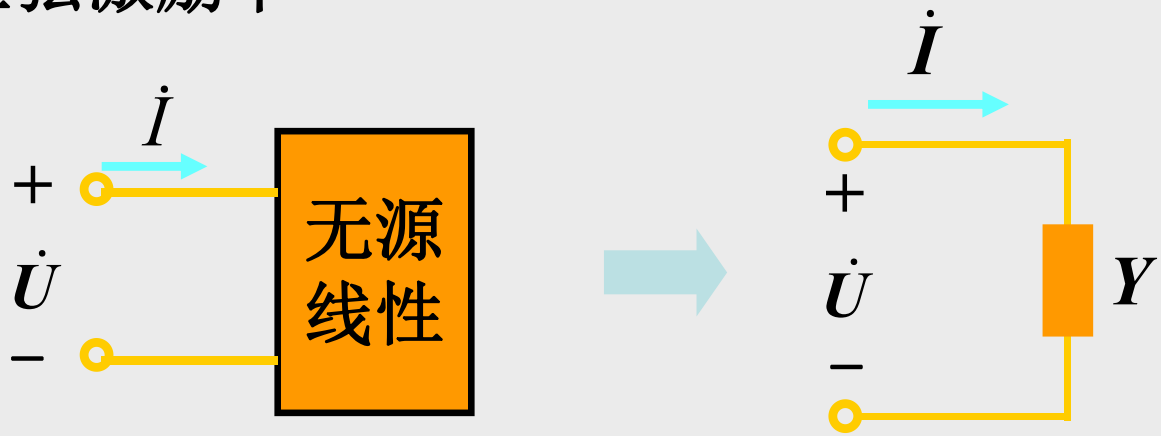


相量图

注 U_L=8.42>U=5，分电压大于总电压。

3. 导纳

正弦激励下



定义导纳 $Y=\dfrac{\dot{I}}{\dot{U}}=|Y|\angle\varphi'$

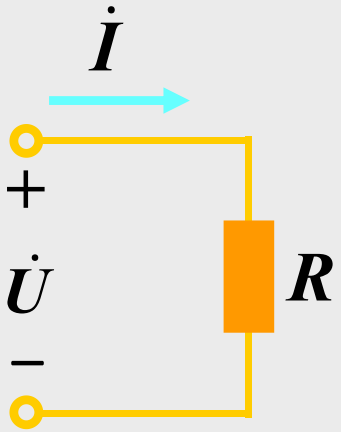
$$|Y|=\frac{I}{U}$$ 导纳模

$$\varphi'=\psi_i-\psi_u$$ 导纳角

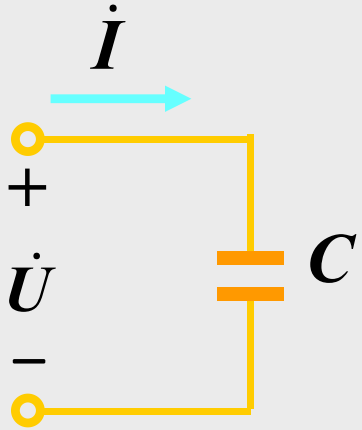
单位：S

对同一二端网络：$Z=\dfrac{1}{Y}$，$Y=\dfrac{1}{Z}$

当无源网络内为单个元件时有：

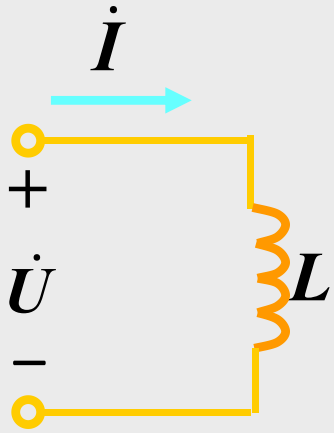


$$Y=\frac{\dot{I}}{\dot{U}}=\frac{1}{R}=G$$



$$Y=\frac{\dot{I}}{\dot{U}}=j\omega C=jB_C$$

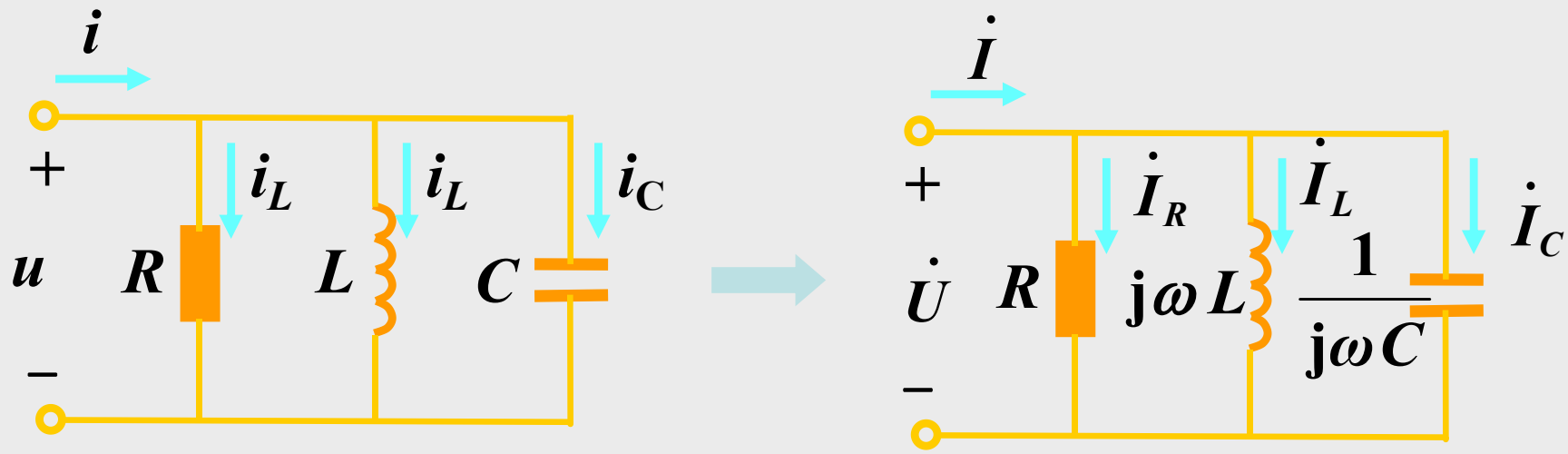
$$B_C=\omega C$$



$$Y=\frac{\dot{I}}{\dot{U}}=1/j\omega L=jB_L \qquad B_L=-1/\omega L$$

Y可以是实数，也可以是虚数

4. *RLC*并联电路



由KCL：
$$\dot{I}=\dot{I}_R+\dot{I}_L+\dot{I}_C=G\dot{U}-\mathrm{j}\frac{1}{\omega L}\dot{U}+\mathrm{j}\omega C\dot{U}$$

$$=(G-\mathrm{j}\frac{1}{\omega L}+\mathrm{j}\omega C)\dot{U}=[G+\mathrm{j}(B_L+B_C)\dot{U}=(G+\mathrm{j}B)\dot{U}$$

$$Y=\frac{\dot{I}}{\dot{U}}=G+j\omega C-j\frac{1}{\omega L}=G+jB=|Y|\angle\varphi'$$

Y— 复导纳；G—电导(导纳的实部)；B—电纳(导纳的虚部)；

$|Y|$—复导纳的模；φ'—导纳角。

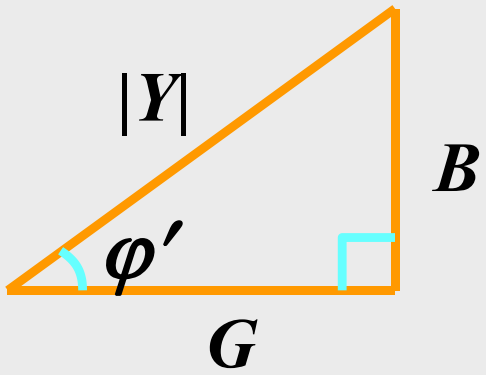
关系：

$$\begin{cases} |Y| = \sqrt{G^2 + B^2} \\ \varphi' = \text{arctg}\dfrac{B}{G} \end{cases}$$

或

$$\begin{cases} G = |Y|\cos\varphi' \\ B = |Y|\sin\varphi' \end{cases} \qquad \begin{cases} |Y| = \dfrac{I}{U} \\ \varphi' = \psi_i - \psi_u \end{cases}$$

导纳三角形



分析 R、L、C 并联电路得出：

（1）$Y=G+j(\omega C-1/\omega L)=|Y|\angle\varphi'$ 为复数，故称复导纳；

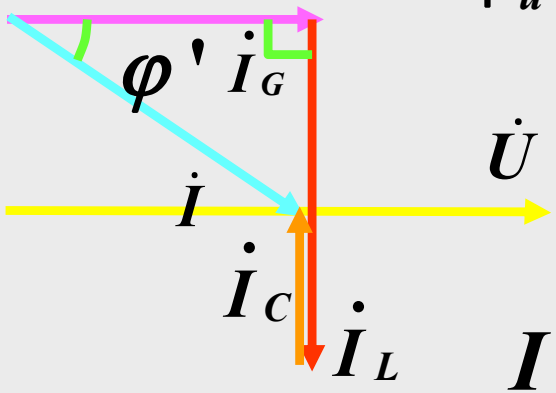
（2）$\omega C > 1/\omega L$ ，$B>0$，$\varphi' >0$，电路为容性，电流超前电压

$\omega C<1/\omega L$ ，$B<0$，$\varphi' <0$，电路为感性，电流落后电压；

$\omega C=1/\omega L$ ，$B=0$，$\varphi' =0$，电路为电阻性，电流与电压同相

（3）相量图：选电压为参考向量，设$\omega C < 1/\omega L$，$\varphi'<0$

$$\psi_u = 0$$

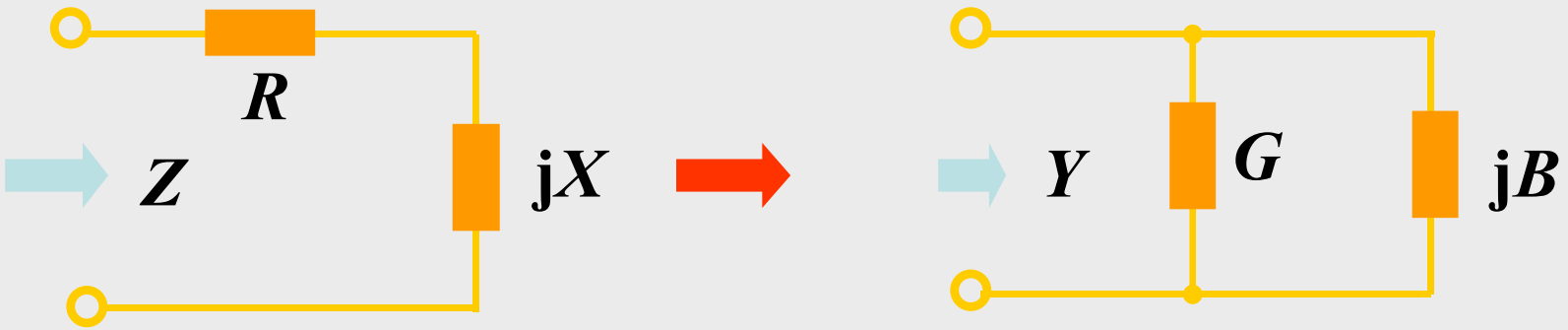


三角形I_R、I_B、I 称为电流三角形，它和导纳三角形相似。即

$$I=\sqrt{I_G^2+I_B^2}=\sqrt{I_G^2+(I_L-I_C)^2}$$

*RLC*并联电路同样会出现分电流大于总电流的现象

5. 复阻抗和复导纳的等效互换



$$Z = R + \mathrm{j}X = |Z| \angle \varphi \quad \Rightarrow \quad Y = G + \mathrm{j}B = |Y| \angle \varphi'$$

$$Y = \frac{1}{Z} = \frac{1}{R+\mathrm{j}X} = \frac{R-\mathrm{j}X}{R^2+X^2} = G + \mathrm{j}B$$

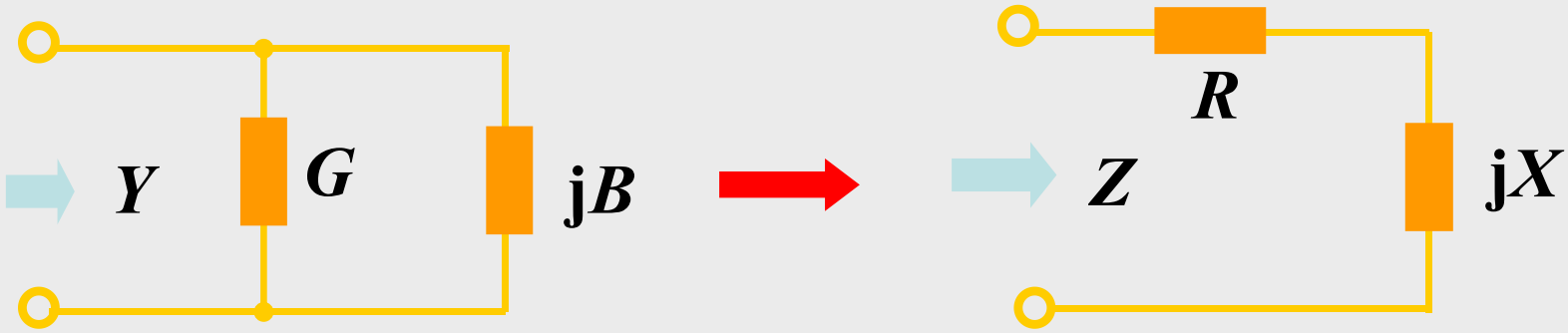
$$\therefore \quad G = \frac{R}{R^2+X^2}, \quad B = \frac{-X}{R^2+X^2} \qquad |Y| = \frac{1}{|Z|}, \quad \varphi' = -\varphi$$

注

一般情况 $G \neq 1/R$ $B \neq 1/X$

若Z为感性，$X>0$，则$B<0$，即仍为感性。

同样，若由*Y*变为*Z*，则有：



$$Y = G + \mathrm{j}B = |Y| \angle \varphi', \quad Z = R + \mathrm{j}X = |Z| \angle \varphi$$

$$Z = \frac{1}{Y} = \frac{1}{G+\mathrm{j}B} = \frac{G-\mathrm{j}B}{G^2+B^2} = R + \mathrm{j}X$$

$$\therefore \quad R = \frac{G}{G^2+B^2}, \quad X = \frac{-B}{G^2+B^2}$$

$$|Y| = \frac{1}{|Z|}, \quad \varphi = -\varphi'$$

例 RL串联电路如图，求在ω＝10^6rad/s时的等效并联电路。

解 RL串联电路的阻抗为：

$$X_L = \omega L = 10^6 \times 0.06 \times 10^{-3} = 60\Omega$$

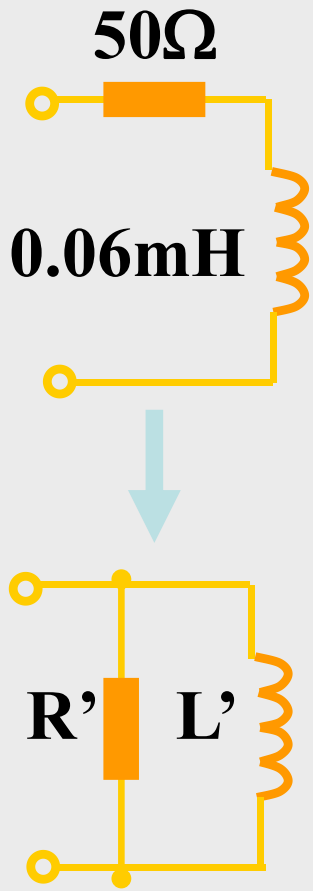
$$Z = R + jX_L = 50 + j60 = 78.1\angle 50.2^0\Omega$$

$$Y = \frac{1}{Z} = \frac{1}{78.1\angle 50.2^0} = 0.0128\angle -50.2^0\Omega$$

$$= 0.0082 - j0.0098 \ \ S$$

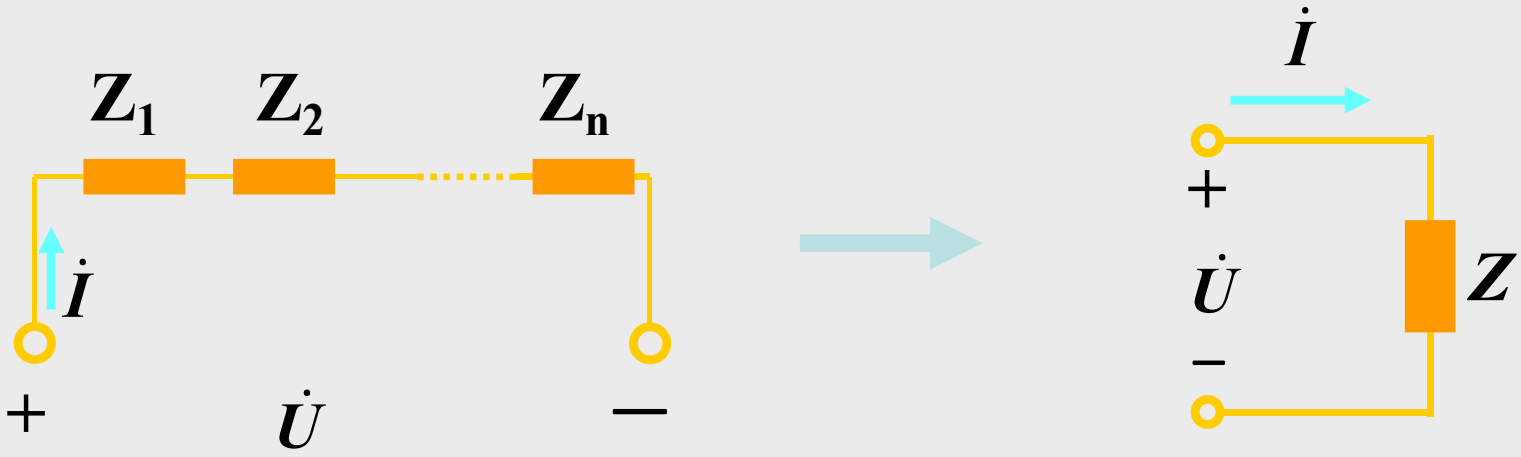
$$R' = \frac{1}{G'} = \frac{1}{0.0082} = 122\Omega$$

$$L' = \frac{1}{0.0098\ \omega} = 0.102\ mH$$



6. 阻抗（导纳）的串联和并联

A. 阻抗的串联



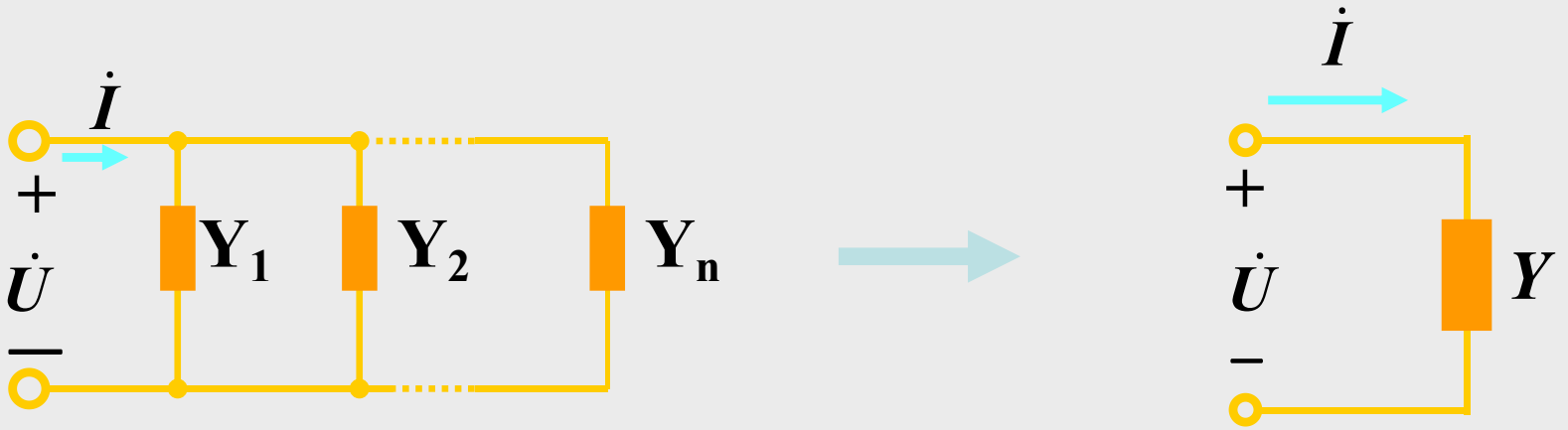
$$\dot{U}=\dot{U}_1+\dot{U}_2+\cdots+\dot{U}_n=\dot{I}(Z_1+Z_2+\cdots+Z_n)=\dot{I}Z$$

$$\rightarrow\quad Z=\sum_{k=1}^{n}Z_k=\sum_{k=1}^{n}(R_k+jX_k)$$

分压公式

$$\dot{U}_i=\frac{Z_i}{Z}\dot{U}$$

B. 导纳的并联



$$\dot{I} = \dot{I}_1 + \dot{I}_2 + \cdots + \dot{I}_n = \dot{U}(Y_1 + Y_2 + \cdots + Y_n) = \dot{U}Y$$

$$\rightarrow \quad Y = \sum_{k=1}^{n} Y_k = \sum_{k=1}^{n} (G_k + jB_k)$$

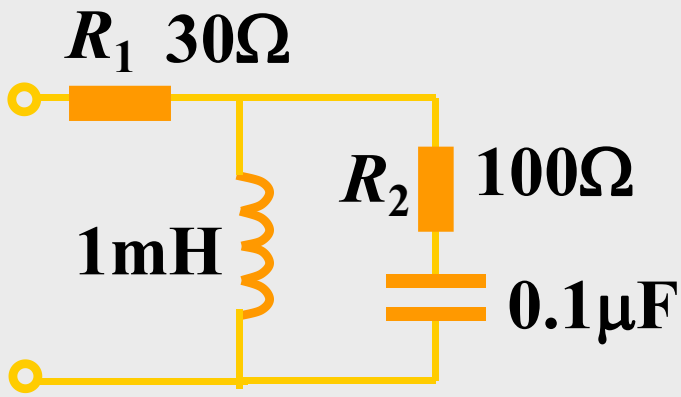
分流公式 $\quad \dot{I}_i = \dfrac{Y_i}{Y}\dot{I}$

两个阻抗Z_1、Z_2的并联等效阻抗为：$\quad Z = \dfrac{Z_1 Z_2}{Z_1 + Z_2}$

例1　求图示电路的等效阻抗，ω＝10^5rad/s 。

解

感抗和容抗为：



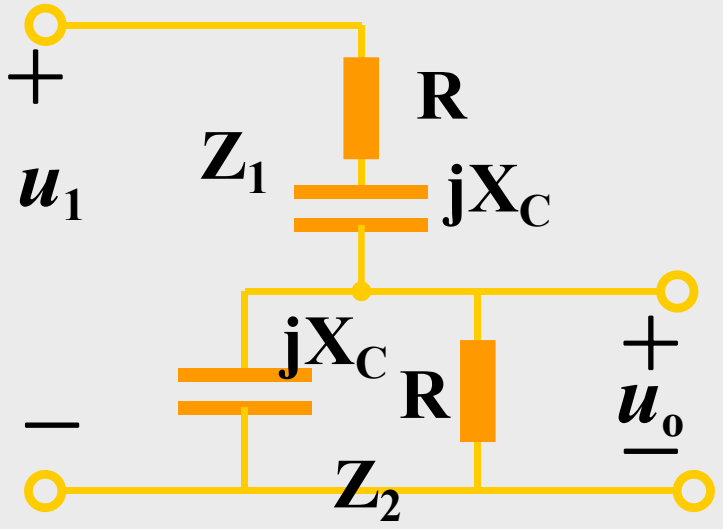
$$X_L = \omega L = 10^5 \times 1 \times 10^{-3} = 100\Omega$$

$$X_C = -\frac{1}{\omega C} = -\frac{1}{10^5 \times 0.1 \times 10^{-6}} = -100\Omega$$

$$Z = R_1 + \frac{jX_L(R_2 + jX_C)}{jX_L + R_2 + jX_C} = 30 + \frac{j100 \times (100 - j100)}{100}$$

$$= 130 + j100\Omega$$

例2　试求u_1和u_0同相位的条件及$\dot{U}_1/\dot{U}_0=?$



解　$Z_1 = R + jX_C$

$$Z_2 = R \| jX_C = \frac{R \times jX_C}{R + jX_C}$$

$$\dot{U}_o = \frac{\dot{U}_1 Z_2}{Z_1 + Z_2} \rightarrow \frac{\dot{U}_1}{\dot{U}_o} = \frac{Z_1 + Z_2}{Z_2} = 1 + \frac{Z_1}{Z_2}$$

$$\frac{Z_1}{Z_2} = \frac{R + jX_C}{jRX_C/(R + jX_C)} = \frac{(R + jX_C)^2}{jRX_C} = \frac{R^2 - X_C^2 + j2RX_C}{jRX_C}$$

$$= 2 - j\frac{R^2 - X_C^2}{RX_C} = \text{实数}$$

$$\rightarrow \quad R = -X_C = \frac{1}{\omega C}$$

$$\frac{\dot{U}_1}{\dot{U}_o} = 1 + 2 = 3$$

8.2 正弦稳态电路的分析

电阻电路与正弦电流电路的分析比较：

电阻电路：

$$\begin{cases} \text{KCL}: \sum i = 0 \\ \text{KVL}: \sum u = 0 \\ \text{元件约束关系}: \ u = Ri \\ \qquad\qquad\qquad \text{或} \ \ i = Gu \end{cases}$$

正弦电路相量分析：

$$\begin{cases} \text{KCL}: \sum \dot{I} = 0 \\ \text{KVL}: \sum \dot{U} = 0 \\ \text{元件约束关系}: \ \dot{U} = Z\dot{I} \\ \qquad\qquad\qquad \text{或} \ \ \dot{I} = Y\dot{U} \end{cases}$$

可见，二者依据的电路定律是相似的。只要作出正弦电流电路的相量模型，便可将电阻电路的分析方法推广应用于正弦稳态的相量分析中。

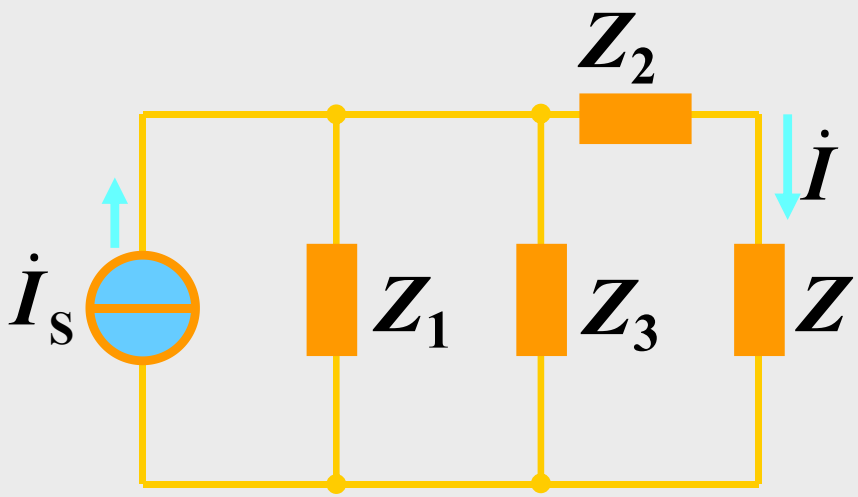
结论

1. 引入相量法，把求正弦稳态电路微分方程的特解问题转化为求解复数代数方程问题。

2. 引入电路的相量模型，不必列写时域微分方程，而直接列写相量形式的代数方程。

3. 引入阻抗以后，可将所有网络定理和方法都应用于交流，直流（f =0)是一个特例。

例1

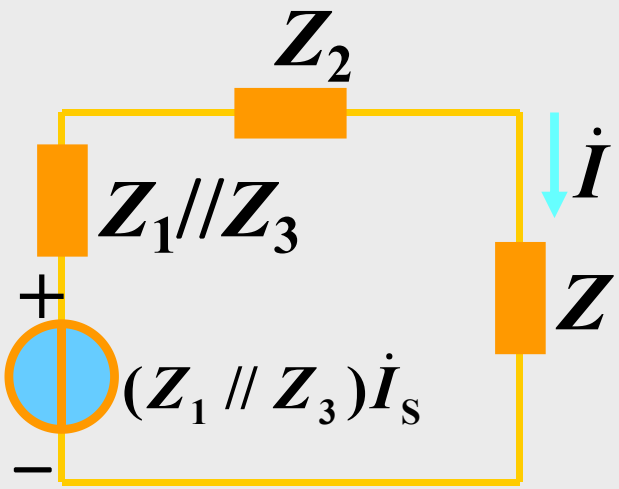


已知：$\dot{I}_S = 4\angle 90^\circ \text{A}$，$Z_1 = Z_2 = -\text{j}30\,\Omega$

$Z_3 = 30\,\Omega$，$Z = 45\,\Omega$

求：$\dot{I}$

解 方法一：电源等效变换



$$Z_1 // Z_3 = \frac{30(-j30)}{30 - j30} = 15 - j15\Omega$$

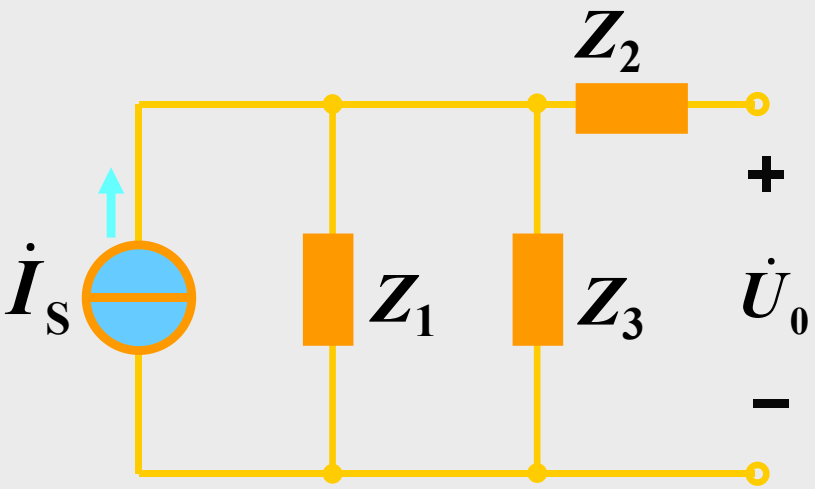
$$\dot{I} = \frac{\dot{I}_S(Z_1 // Z_3)}{Z_1 // Z_3 + Z_2 + Z}$$

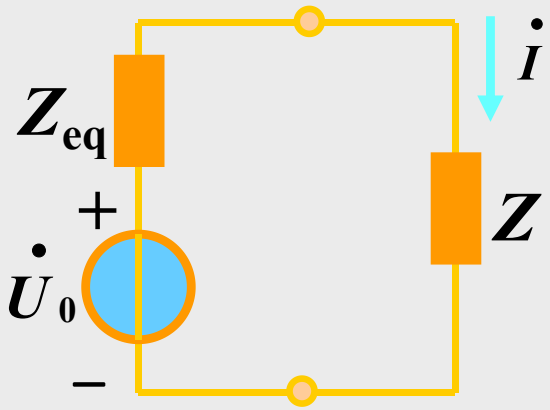
$$= \frac{\mathrm{j}4(15 - \mathrm{j}15)}{15 - \mathrm{j}15 - \mathrm{j}30 + 45}$$

$$= \frac{5.657\angle 45^\circ}{5\angle -36.9^\circ}$$

$$= 1.13\angle 81.9^\circ \ \mathrm{A}$$

方法二：戴维南等效变换





求开路电压：

$$\dot{U}_0 = \dot{I}_S(Z_1 // Z_3)$$

$$= 84.86\angle 45^o \text{ V}$$

求等效电阻：

$$Z_{eq} = Z_1 // Z_3 + Z_2$$

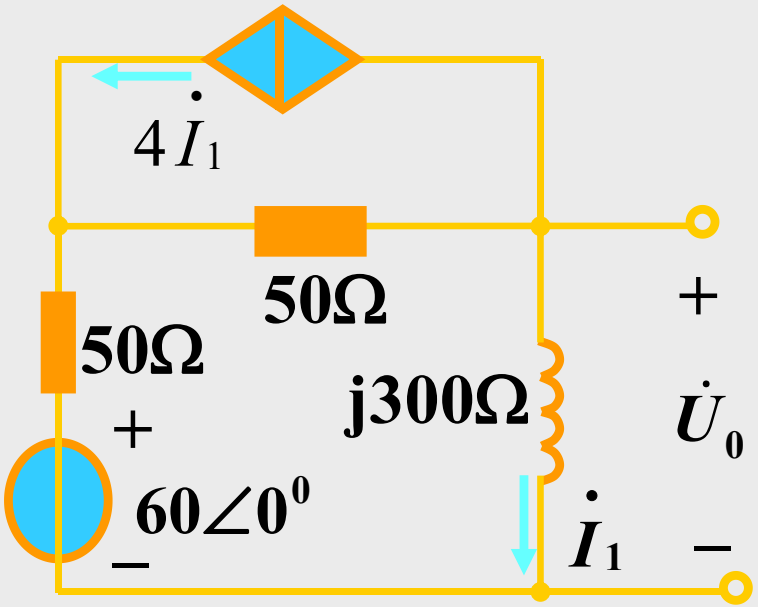
$$= 15 - j45\Omega$$

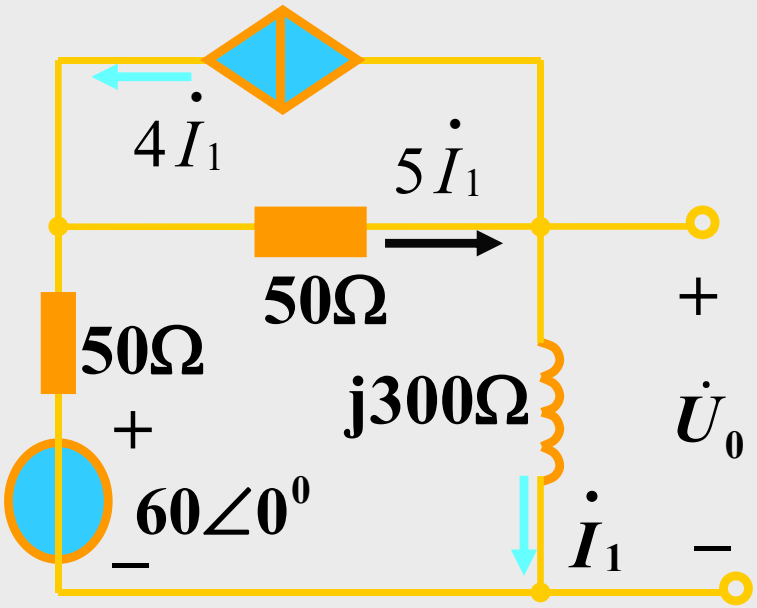
$$\dot{I} = \frac{\dot{U}_0}{Z_0 + Z} = \frac{84.86\angle 45^\circ}{15 - j45 + 45}$$

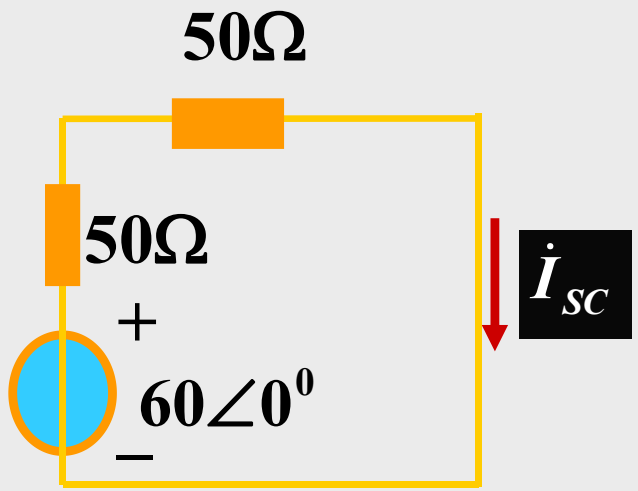
$$= 1.13\angle 81.9^o \text{ A}$$

例2

求图示电路的戴维南等效电路。







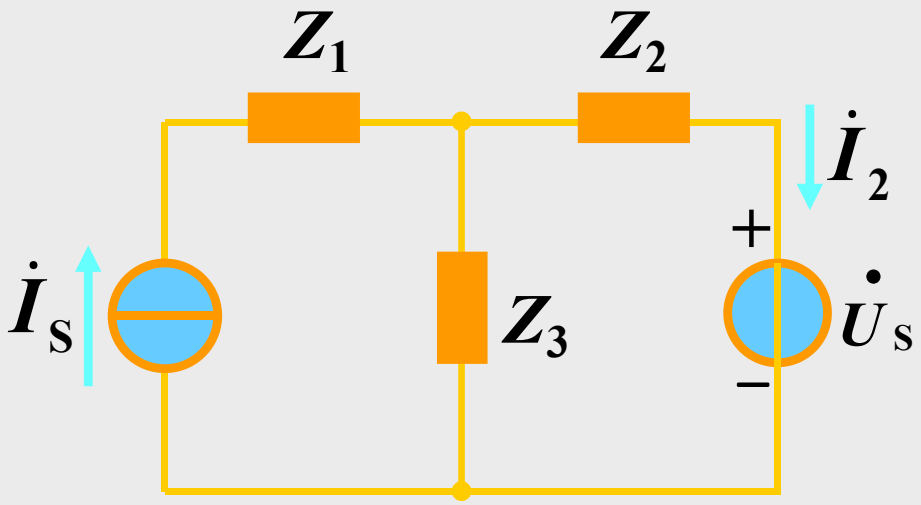
解 $\dot{U}_o = -5\dot{I}_1 * 50 - \dot{I}_1 * 50 + 60 = -300\dot{I}_1 + 60 = -300\dfrac{\dot{U}_0}{j300} + 60$

$$\rightarrow \quad \dot{U}_o = \frac{60}{1-j} = 30\sqrt{2}\angle 45^0$$

求短路电流： $\dot{I}_{SC} = 60/100 = 0.6\angle 0^0$

$$\rightarrow \quad Z_{eq} = \frac{\dot{U}_0}{\dot{I}_{SC}} = \frac{30\sqrt{2}\angle 45^0}{0.6} = 50\sqrt{2}\angle 45^0$$

例3 用叠加定理计算电流 $\dot{I}_2$



已知：$\dot{U}_S = 100\angle 45^\circ \text{ V},$

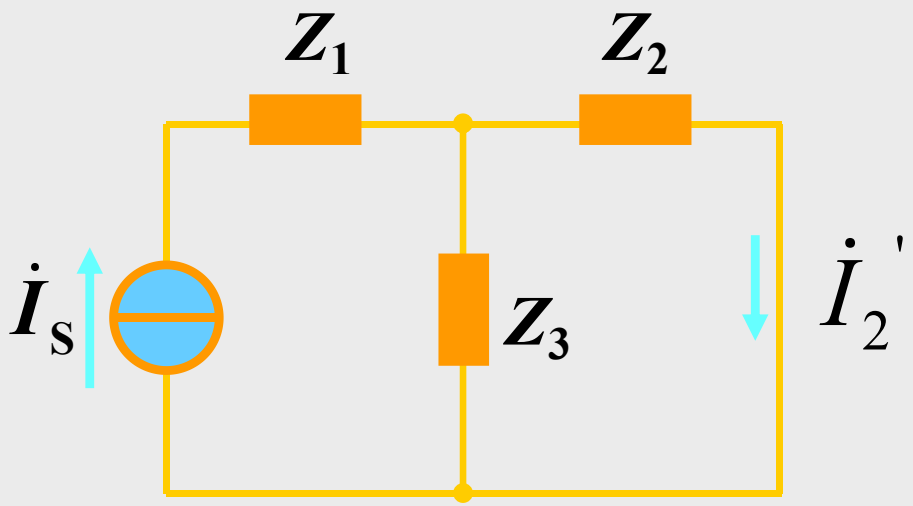
$$\dot{I}_S = 4\angle 0^\circ \text{ A},$$

$$Z_1 = Z_3 = 50\angle 30^\circ \Omega,$$

$$Z_3 = 50\angle -30^\circ \Omega .$$

解

(1) $\dot{I}_S$ 单独作用($\dot{U}_S$ 短路)：

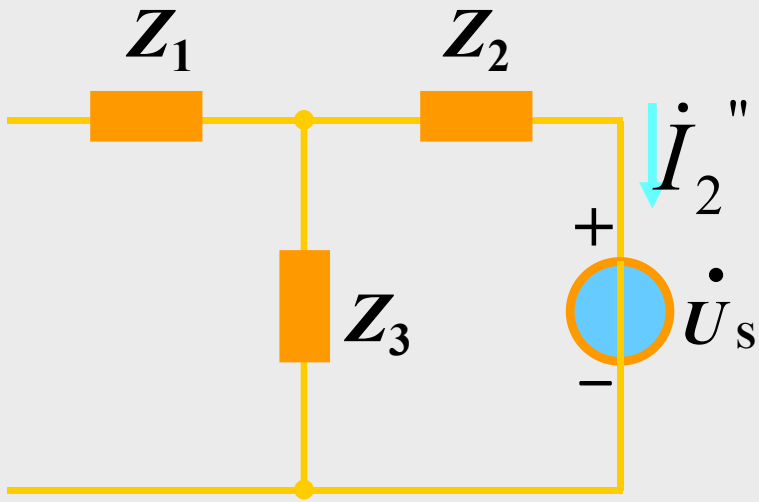


$$\dot{I}_2' = \dot{I}_S \frac{Z_3}{Z_2 + Z_3}$$

$$= 4\angle 0^o \times \frac{50\angle 30^o}{50\angle -30^o + 50\angle 30^o}$$

$$= \frac{200\angle 30^o}{50\sqrt{3}} = 2.31\angle 30^o \text{ A}$$

(2) $\dot{U}_S$ 单独作用($\dot{I}_S$ 开路)：



$$\dot{I}_2'' = -\frac{\dot{U}_S}{Z_2 + Z_3}$$

$$= \frac{-100\angle 45^\circ}{50\sqrt{3}} = 1.155\angle -135^\circ \text{ A}$$

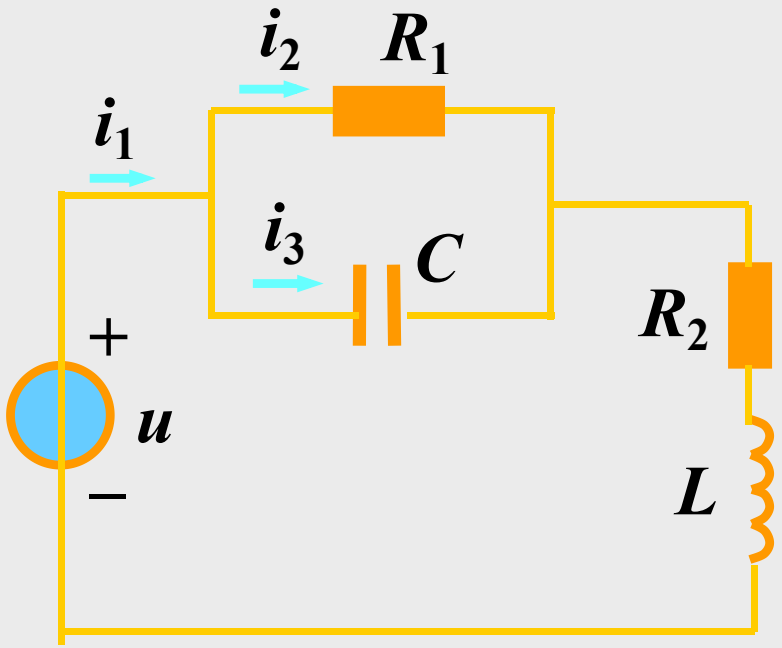
$$\dot{I}_2 = \dot{I}_2' + \dot{I}_2''$$

$$= 2.31\angle 30^\circ + 1.155\angle -135^\circ = 1.23\angle -15.9^\circ \text{ A}$$

例4

已知：$R_1=1000\Omega$，$R_2=10\Omega$，$L=500mH$，$C=10\mu F$，

$U=100V$，$\omega=314rad/s$，求：各支路电流。



解

画出电路的相量模型

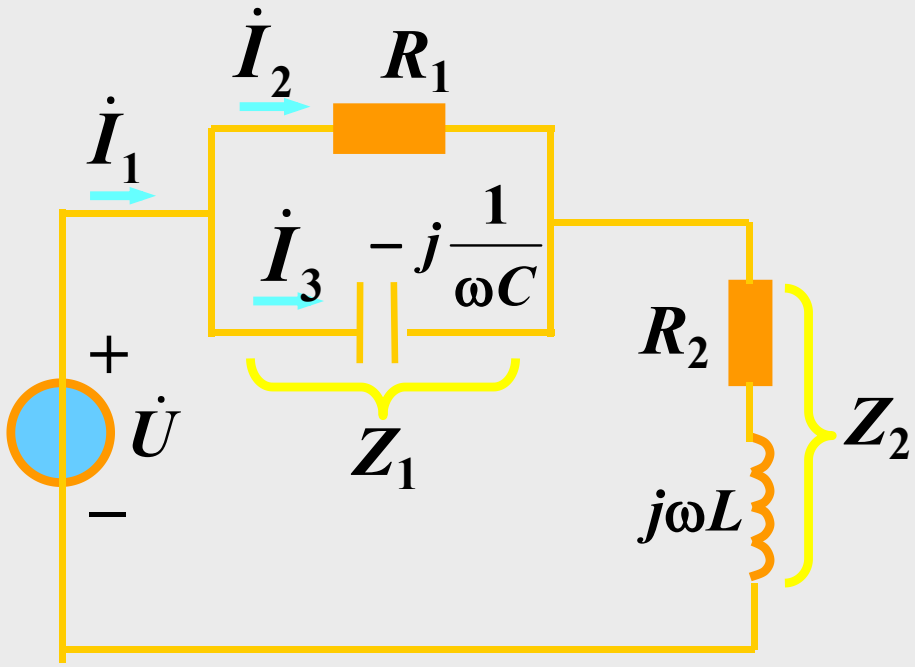
$$Z_1 = \frac{R_1(-j\,1/\omega C)}{R_1 - j\,1/\omega C}$$

$$= \frac{1000\times(-j318.47)}{1000 - j318.47}$$

$$= \frac{318.47\times10^3\angle -90^\circ}{1049.5\angle -17.7^\circ}$$

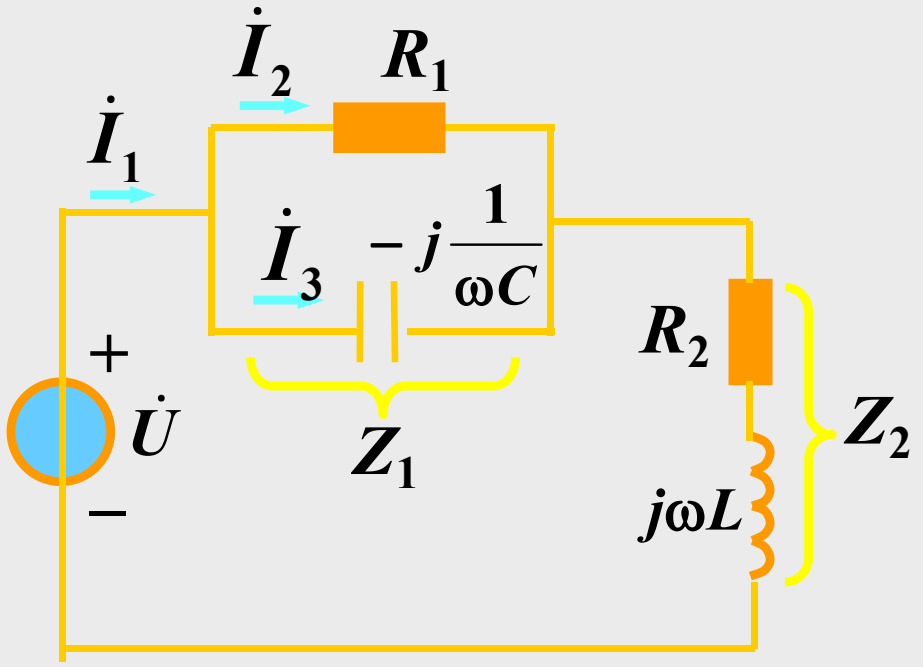
$$= 303.45\angle -72.3^\circ$$

$$= 92.11 - j289.13\,\Omega$$



$$Z_2 = R_2 + j\omega L = 10 + j157\,\Omega$$

$$Z = Z_1 + Z_2$$
$$= 92.11 - j289.13 + 10 + j157$$
$$= 102.11 - j132.13$$
$$= 166.99\angle -52.3^\circ\ \Omega$$



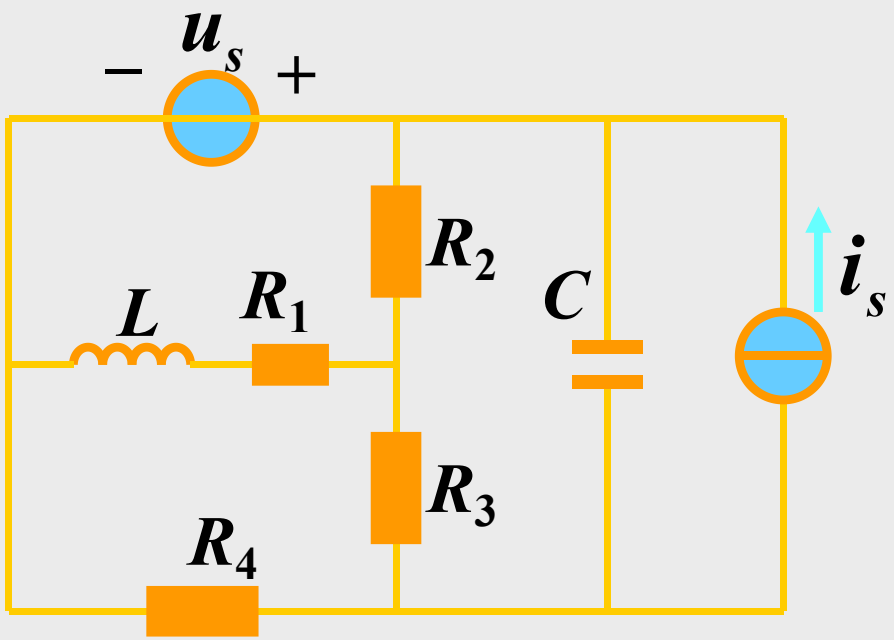
$$\dot{I}_1 = \frac{\dot{U}}{Z} = \frac{100\angle 0^\circ}{166.99\angle -52.3^\circ} = 0.6\angle 52.3^\circ\ A$$

$$\dot{I}_2 = \frac{-j\,1/\omega C}{R_1 - j\frac{1}{\omega C}}\dot{I}_1 = \frac{-j318.47}{1049.5\angle -17.7^\circ} \times 0.6\angle 52.3^\circ = 0.181\angle -20^\circ\ A$$

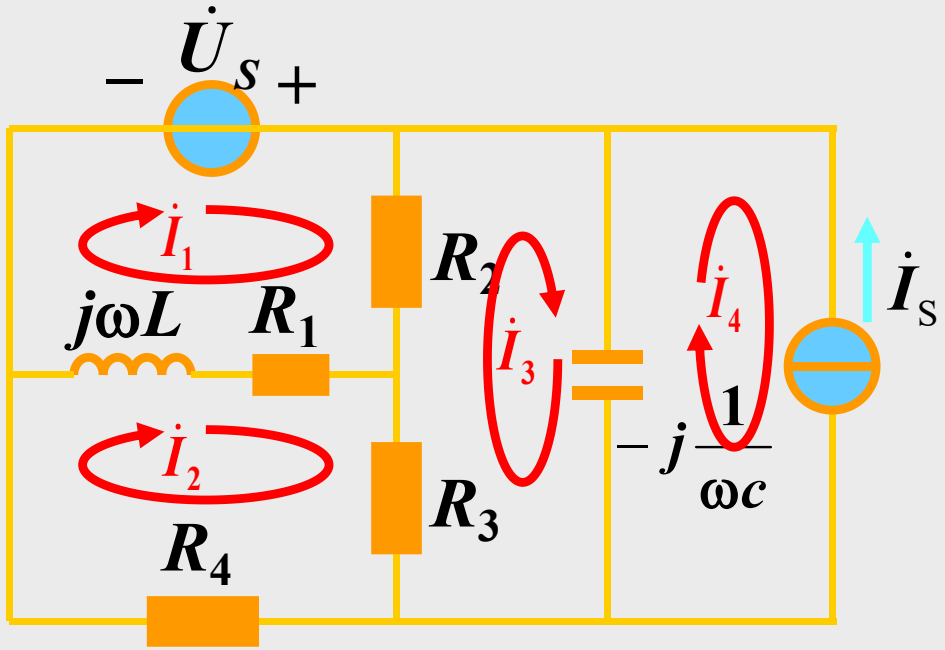
$$\dot{I}_3 = \frac{R_1}{R_1 - j\frac{1}{\omega C}}\dot{I}_1 = \frac{1000}{1049.5\angle -17.7^\circ} \times 0.6\angle 52.3^\circ = 0.57\angle 70^\circ\ A$$

例5

列写电路的回路电流方程和节点电压方程

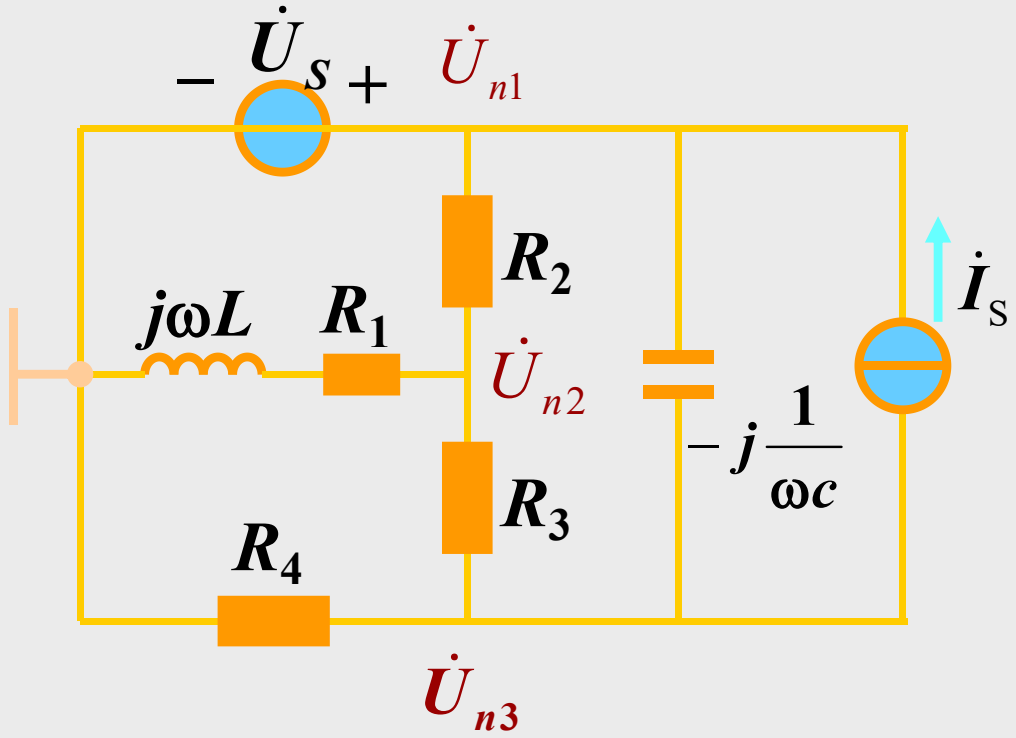


解



回路法:

$$\begin{cases} (R_1+R_2+j\omega L)\dot{I}_1-(R_1+j\omega L)\dot{I}_2-R_2\dot{I}_3=\dot{U}_S \\ (R_1+R_3+R_4+j\omega L)\dot{I}_2-(R_1+j\omega L)\dot{I}_1-R_3\dot{I}_3=0 \\ (R_2+R_3-j\dfrac{1}{\omega C})\dot{I}_3-R_2\dot{I}_1-R_3\dot{I}_2-j\dfrac{1}{\omega C}\dot{I}_4=0 \\ \dot{I}_4=-\dot{I}_S \end{cases}$$

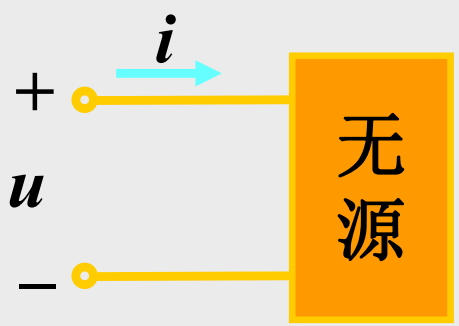


节点法:

$$\begin{cases} \dot{U}_{n1} = \dot{U}_S \\ (\dfrac{1}{R_1 + j\omega L} + \dfrac{1}{R_2} + \dfrac{1}{R_3})\dot{U}_{n2} - \dfrac{1}{R_2}\dot{U}_{n1} - \dfrac{1}{R_3}\dot{U}_{n3} = 0 \\ (\dfrac{1}{R_3} + \dfrac{1}{R_4} + j\omega C)\dot{U}_{n3} - \dfrac{1}{R_3}\dot{U}_{n2} - j\omega C\dot{U}_{n1} = -\dot{I}_S \end{cases}$$

8.3 正弦稳态电路的功率

无源一端口网络吸收的功率



$$u(t)=\sqrt{2}U\cos\omega t$$

$$i(t)=\sqrt{2}I\cos(\omega t-\varphi)$$

φ为u和i的相位差$\varphi=\Psi_u-\Psi_i$

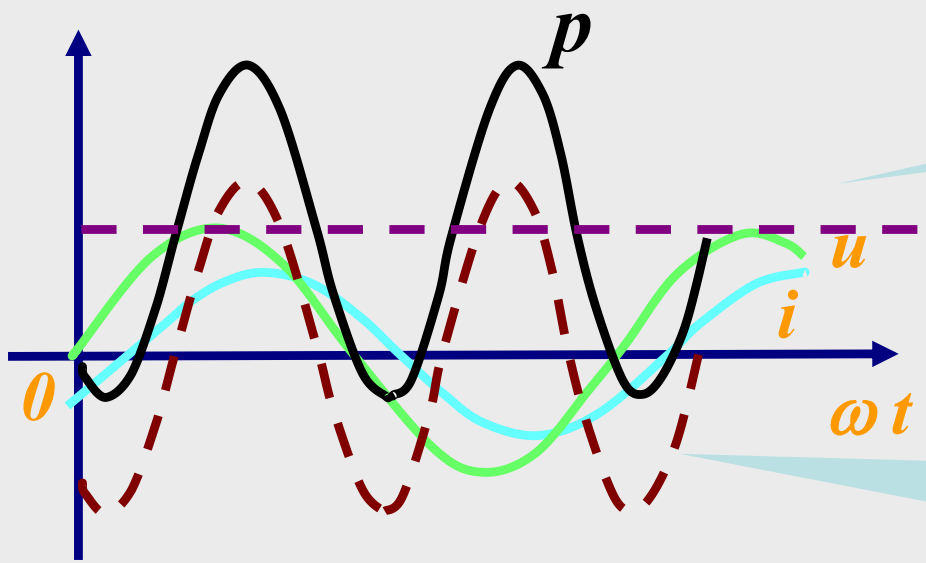
1. 瞬时功率 (*instantaneous power*)

$$p(t)=ui=\sqrt{2}U\cos\omega t\cdot\sqrt{2}I\cos(\omega t-\varphi)$$

$$=UI[\cos\varphi+\cos(2\omega t-\varphi)]$$

$$2\cos\alpha\cos\beta=\cos(\alpha+\beta)+\cos(\alpha-\beta)$$

$$p(t) = UI[\cos\varphi + \cos(2\omega t - \varphi)]$$



$UI\cos\varphi$ 恒定分量。

$UI\cos(2\omega t - \varphi)$ 为正弦分量。

- p有时为正，有时为负；
- $p>0$, 电路吸收功率；
- $p<0$，电路发出功率；

2. 平均功率 (*average power*)

$$P=\frac{1}{T}\int_0^T p\mathrm{d}t=\frac{1}{T}\int_0^T[UI\cos\varphi+UI\cos(\omega t+\varphi)]\mathrm{d}t$$

$$=UI\cos\varphi$$

$$\boldsymbol{P=UI\cos\varphi}$$

P 的单位：W（瓦）

$\varphi=\psi_u-\psi_i$：功率因数角。对无源网络，为其等效阻抗的阻抗角。

$\cos\varphi$：功率因数。

$\cos\varphi$ { 1，纯电阻；0，纯电抗 }　　一般地，有 $0\le|\cos\varphi|\le1$

$\varphi>0$，$[0,\pi]$，感性　　$\varphi<0$，$[-\pi,0]$，容性

$$P=UI\cos\varphi=I^2|Z|\cos\varphi=RI^2$$

平均功率实际上是电阻消耗的功率，亦称为有功功率。表示电路实际消耗的功率，它不仅与电压电流有效值有关，而且与 $\cos\varphi$有关，这是交流和直流的很大区别，主要是由于电压、电流存在相位差。

3. 无功功率 (*reactive power*) *Q*

$$Q \overset{def}{=} UI\sin\varphi = I^2|Z|\sin\varphi = XI^2$$

表示交换功率的大小，单位：**var (乏)**

Q*>0**，表示网络吸收无功功率；Q*<0**，表示网络发出无功功率

Q 的大小反映网络与外电路交换功率的大小，由储能元件***L***、***C*** 的性质决定

4. 视在功率*S*

$$S \overset{\text{def}}{=} UI \qquad \text{单位：VA (伏安)}$$

反映电气设备的容量。

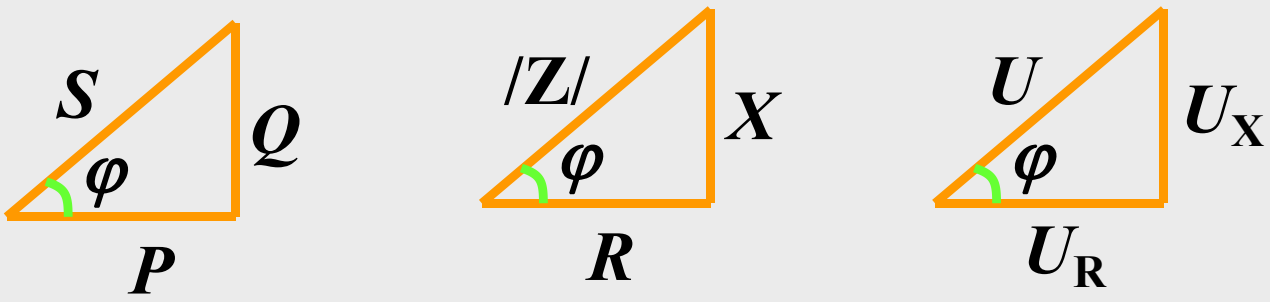
有功，无功，视在功率的关系：

有功功率：$P=UI\cos\varphi$　　单位：W

无功功率：$Q=UI\sin\varphi$　　单位：var

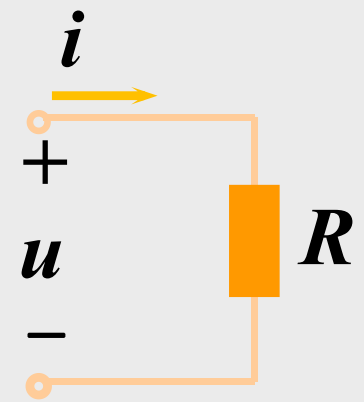
视在功率：$S=UI$　　单位：VA

$$S=\sqrt{P^2+Q^2}$$



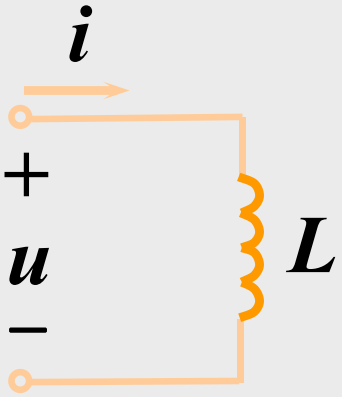
功率三角形　　阻抗三角形　　电压三角形

5. R、L、C元件的有功功率和无功功率



$$P_R = UI\cos\varphi = UI\cos 0° = UI = I^2R = U^2/R$$

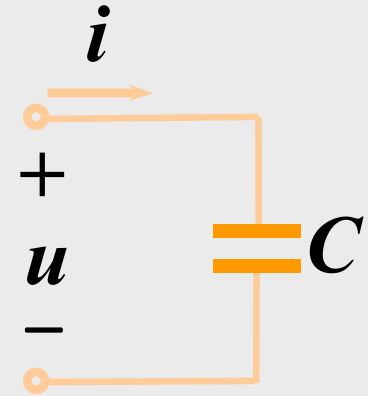
$$Q_R = UI\sin\varphi = UI\sin 0° = 0$$



$$P_L = UI\cos\varphi = UI\cos 90° = 0$$

$$Q_L = UI\sin\varphi = UI\sin 90° = UI = I^2X_L$$

吸收无功为正

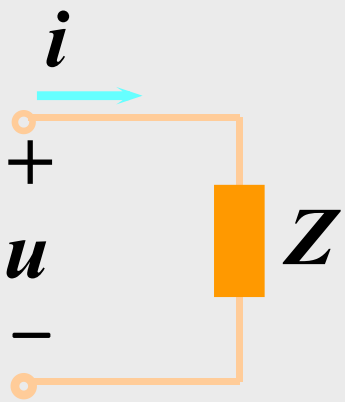


$$P_C = UI\cos\varphi = UI\cos(-90°) = 0$$

$$Q_C = UI\sin\varphi = UI\sin(-90°) = -UI = -I^2X_C$$

吸收无功为负　(发出无功)

任意阻抗的功率计算：



$$P_Z = UI\cos\varphi = I^2|Z|\cos\varphi = I^2R$$

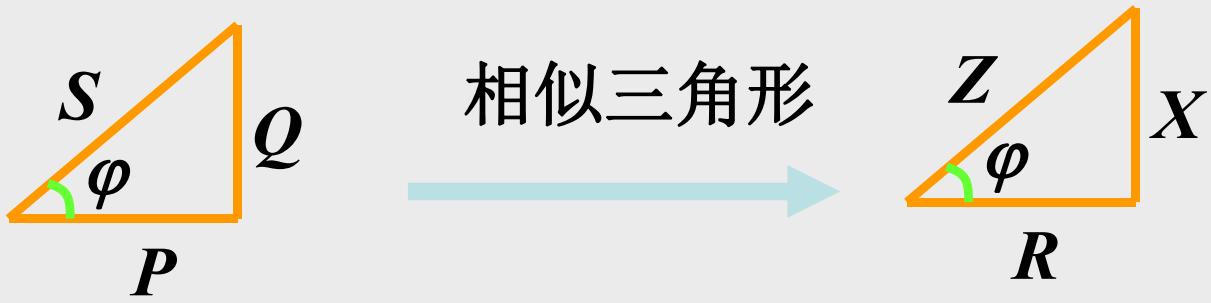
$$Q_Z = UI\sin\varphi = I^2|Z|\sin\varphi = I^2X$$

$$= I^2(X_L + X_C) = Q_L + Q_C$$

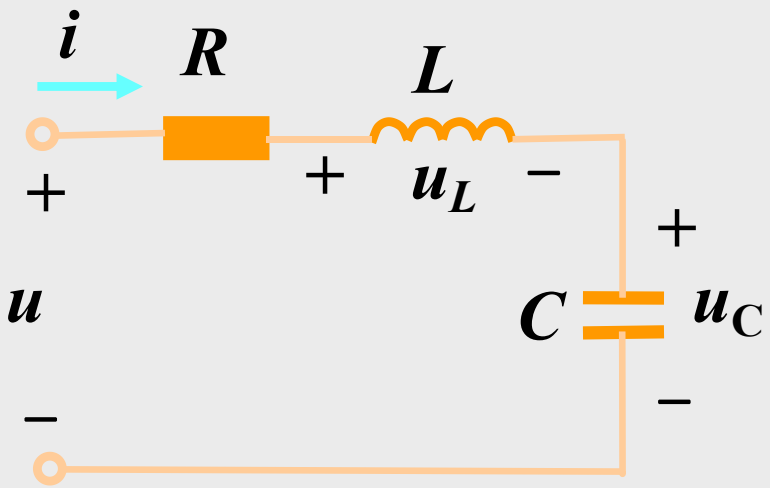
$Q_L = I^2X_L > 0$　　吸收无功为正

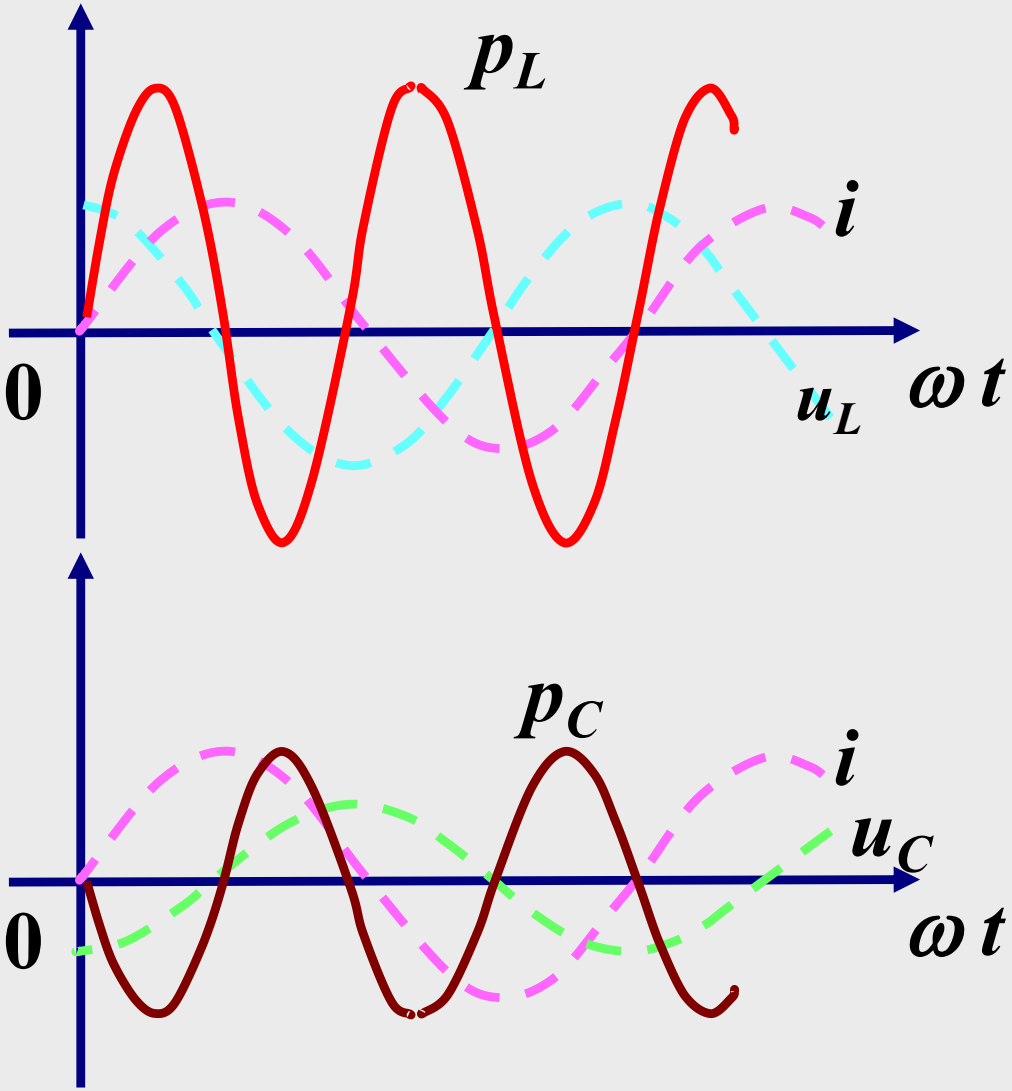
$Q_C = I^2X_C < 0$　　吸收无功为负

$$S = \sqrt{P^2 + Q^2} = I^2\sqrt{R^2 + X^2} = I^2|Z|$$



电感、电容的无功补偿作用



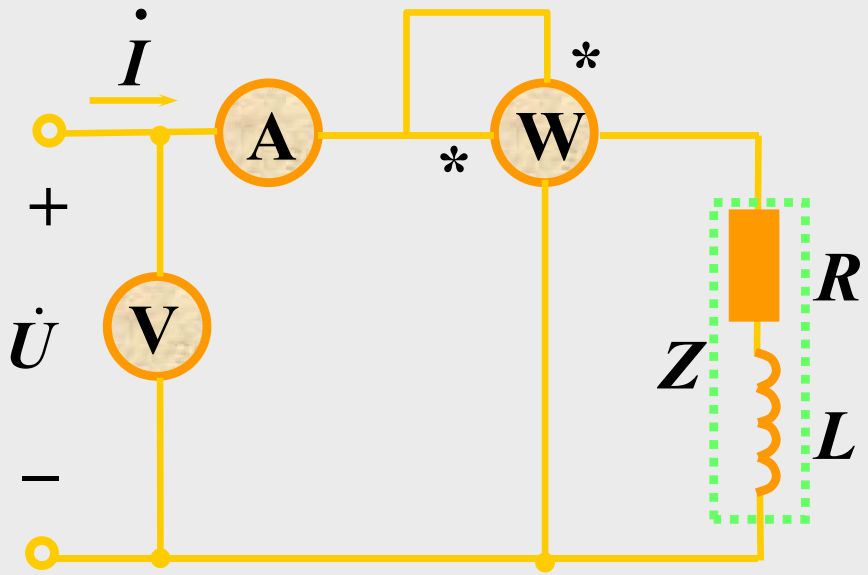


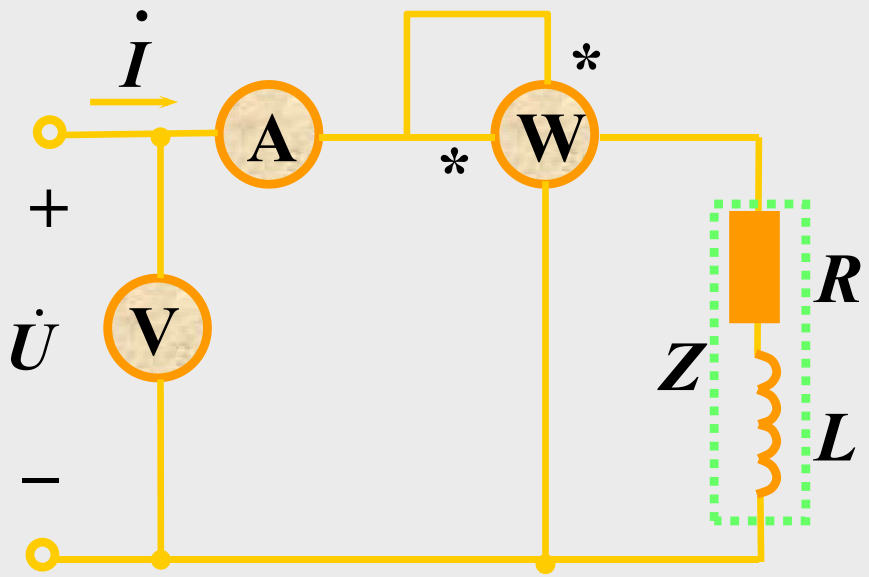
当*L*发出功率时，*C*刚好吸收功率。因此，*L*、*C*的无功具有互相补偿的作用。

例1

三表法测线圈参数。

已知f=50Hz，且测得U=50V，I=1A，P=30W。





解 方法一

$$S = UI = 50 \times 1 = 50VA$$

$$Q = \sqrt{S^2 - P^2} = \sqrt{50^2 - 30^2} = 40VAR$$

$$R = \frac{P}{I^2} = \frac{30}{1} = 30\Omega \qquad X_L = \frac{Q}{I^2} = \frac{40}{1} = 40\Omega$$

$$\rightarrow \quad L = \frac{X_L}{\omega} = \frac{40}{100\pi} = 0.127H$$

方法二　　$P = I^2 R$　　$\therefore R = \dfrac{P}{I^2} = \dfrac{30}{1^2} = 30\Omega$

$|Z| = \dfrac{U}{I} = \dfrac{50}{1} = 50\Omega$　　又　　$|Z| = \sqrt{R^2 + (\omega L)^2}$

$\rightarrow$　$L = \dfrac{1}{\omega}\sqrt{|Z|^2 - R^2} = \dfrac{1}{314}\sqrt{50^2 - 30^2} = \dfrac{40}{314} = 0.127\text{H}$

方法三　　$P = UI\cos\varphi$　$\rightarrow$　$\cos\varphi = \dfrac{P}{UI} = \dfrac{30}{50 \times 1} = 0.6$

$|Z| = \dfrac{U}{I} = \dfrac{50}{1} = 50\Omega$

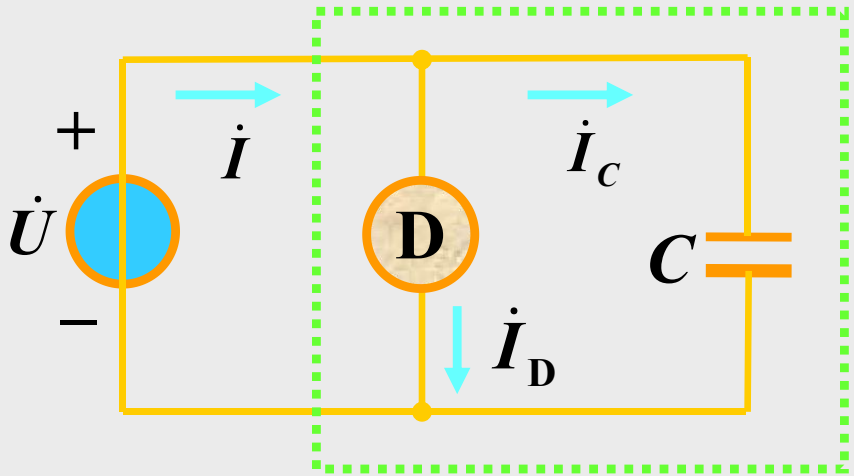
$R = |Z|\cos\varphi = 50 \times 0.6 = 30\Omega$

$X_L = |Z|\sin\varphi = 50 \times 0.8 = 40\Omega$

例2

已知：电动机的 P_D=1000W，电感性，功率因数为0.8，U=220，f=50Hz，C=30μF。

求负载电路的功率因数。



解

$$I_{\mathrm{D}}=\frac{P_{\mathrm{D}}}{U\cos\varphi_{\mathrm{D}}}=\frac{1000}{220\times 0.8}=5.68\mathrm{A}$$

$$\because\ \cos\varphi_{\mathrm{D}}=0.8(\text{电流滞后}),\quad \therefore\ \varphi_{\mathrm{D}}=36.8^{\circ}$$

设 $\dot{U}=220\angle 0^{\circ}$

$$\dot{I}_{\mathrm{D}}=5.68\angle -36.8^{\circ}\ ,\ \dot{I}_{C}=220\angle 0^{\circ}\cdot \mathrm{j}\omega C=\mathrm{j}2.08$$

$$\dot{I}=\dot{I}_{\mathrm{D}}+\dot{I}_{C}=4.55-\mathrm{j}1.32=4.74\angle -16.2^{\circ}$$

$$\cos\varphi=\cos(-16.2^{\circ})=0.96$$

6. 功率因数提高

$$P=UI\cos\varphi$$

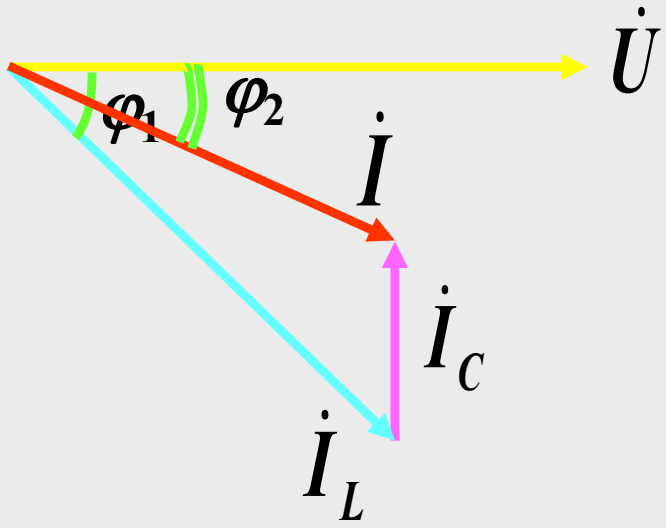
功率因数低带来的问题：

（1）设备不能充分利用，电流到了额定值，但功率还有容量；

（2）当输出相同的有功功率时，线路上电流大，$I=P/(U\cos\varphi)$，线路损耗大。

解决办法：

在电感性负载上并联电容，提高功率因数（改进自身设备）。



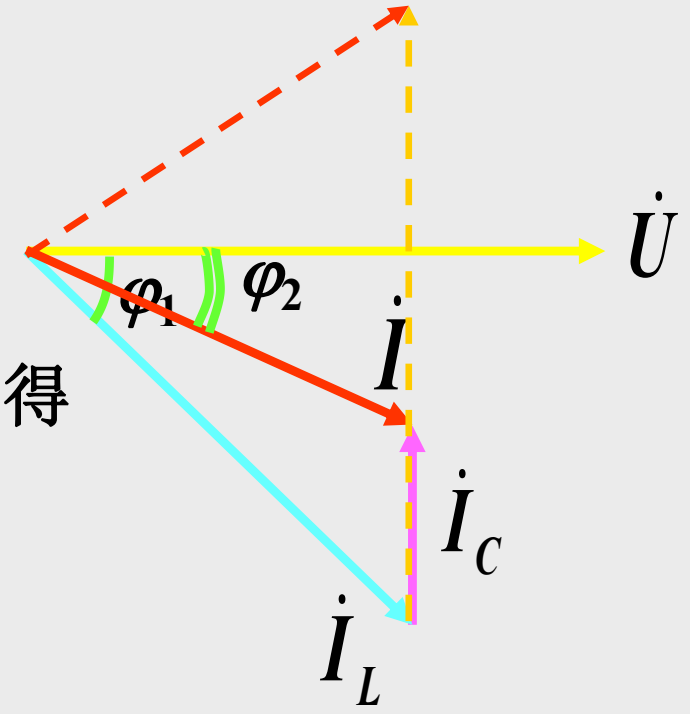
特点：

并联电容后，原负载的电压和电流不变，吸收的有功功率和无功功率不变，即：负载的工作状态不变。

但电路的功率因数提高了。

并联电容的确定：

$$I_C = I_L \sin\varphi_1 - I\sin\varphi_2$$



将 $I = \dfrac{P}{U\cos\varphi_2}$ ， $I_L = \dfrac{P}{U\cos\varphi_1}$ 代入得

$$I_C = \omega CU = \frac{P}{U}(\mathrm{tg}\varphi_1 - \mathrm{tg}\varphi_2)$$

$$\rightarrow \quad C = \frac{P}{\omega U^2}(\mathrm{tg}\varphi_1 - \mathrm{tg}\varphi_2)$$

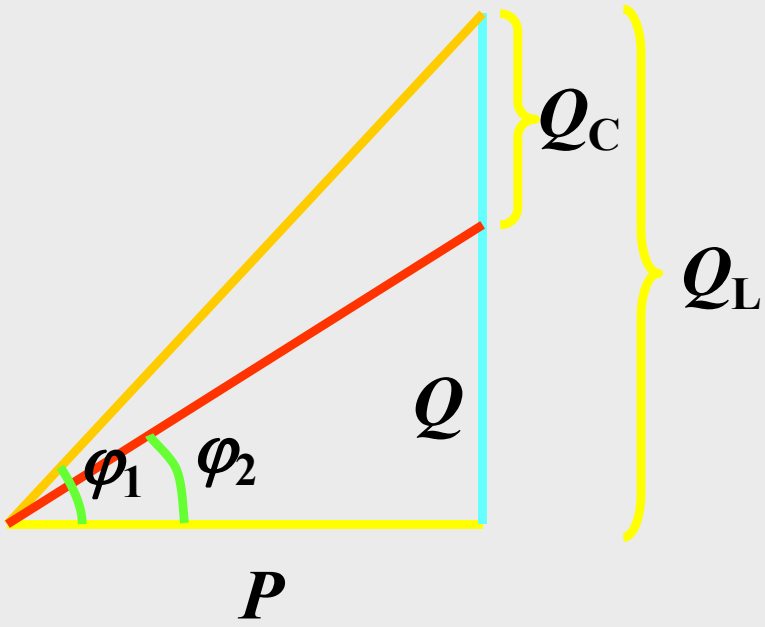
补偿容量不同

欠

全——不要求(电容设备投资增加, 经济效果不明显)

过——使功率因数又由高变低(性质不同)

并联电容也可以用功率三角形确定：



$$\left|Q_C\right| = \left|Q_L - Q\right| = P(\mathrm{tg}\varphi_1 - \mathrm{tg}\varphi_2)$$

$$\left|Q_C\right| = \omega C U^2$$

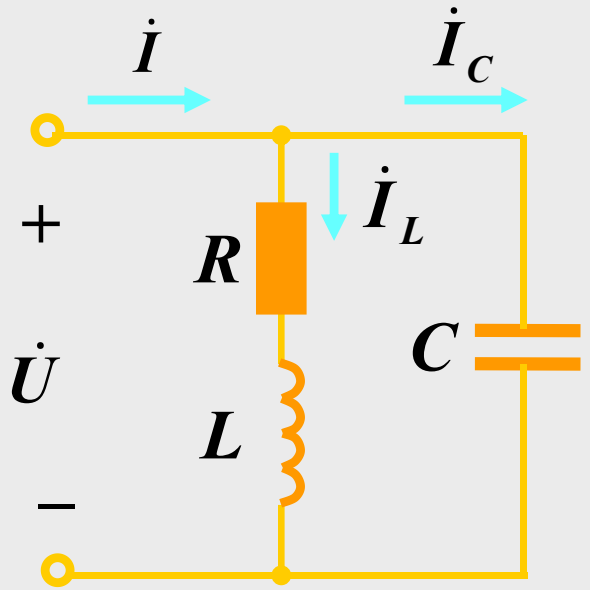
$$\therefore \quad C = \frac{P}{\omega U^2}(\mathrm{tg}\varphi_1 - \mathrm{tg}\varphi_2)$$

从功率这个角度来看：

并联电容后，电源向负载输送的有功功率不变，但是电源向负载输送的无功功率减少了，减少的这部分无功就由电容发出来补偿，使感性负载吸收的无功不变，而功率因数得到改善。

例

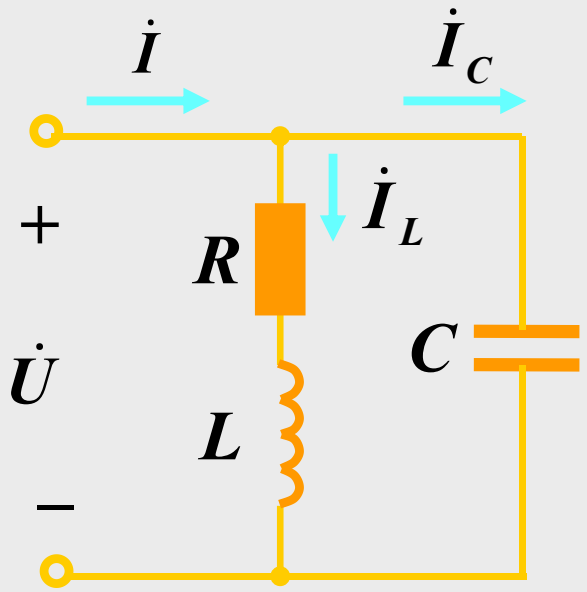
已知：f=50Hz, U=220V, P=10kW, $\cos\varphi_1$=0.6，要使功率因数提高到0.9 , 求并联电容C，并联前后电路的总电流各为多大？



解

$$\cos\varphi_1 = 0.6 \Rightarrow \varphi_1 = 53.13^o$$

$$\cos\varphi_2 = 0.9 \Rightarrow \varphi_2 = 25.84^o$$



$$C = \frac{P}{\omega U^2}(\mathrm{tg}\varphi_1 - \mathrm{tg}\varphi_2)$$

$$= \frac{10\times10^3}{314\times220^2}(\mathrm{tg}53.13^\circ - \mathrm{tg}25.84^\circ) = 557\mu\,\mathrm{F}$$

未并电容时：$I = I_L = \dfrac{P}{U\cos\varphi_1} = \dfrac{10\times10^3}{220\times0.6} = 75.8A$

并联电容后：$I = \dfrac{P}{U\cos\varphi_2} = \dfrac{10\times10^3}{220\times0.9} = 50.5A$

若要使功率因数从0.9再提高到0.95，试问还应增加多少并联电容，此时电路的总电流是多大？

解 $\cos\varphi_1 = 0.9 \Rightarrow \varphi_1 = 25.84^{o}$ $\cos\varphi_2 = 0.95 \Rightarrow \varphi_2 = 18.19^{o}$

$$C = \frac{P}{\omega U^2}(\mathrm{tg}\varphi_1 - \mathrm{tg}\varphi_2)$$

$$= \frac{10\times10^3}{314\times220^2}(\mathrm{tg}25.84^\circ - \mathrm{tg}18.19^\circ) = 103\mu\,\mathrm{F}$$

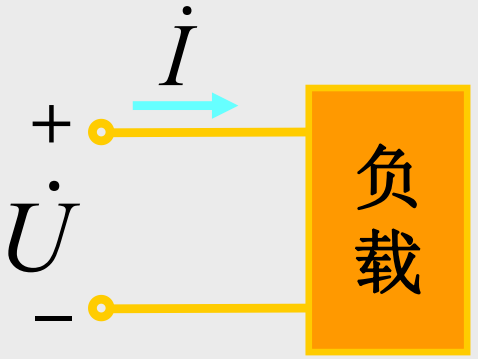
$$I = \frac{10\times10^3}{220\times0.95} = 47.8A$$

可见，功率因数提高后，线路上总电流减少，但继续提高功率因数所需电容很大，增加成本，总电流减小却不明显。因此一般将功率因数提高到0.9即可。

8.4 复功率

1. 复功率

为了用相量$\dot{U}$和$\dot{I}$来计算功率，引入“复功率”



定义：$\overline{S}=\dot{U}\dot{I}^{*}$ 单位 VA

$$\overline{S}=UI\angle(\Psi_u-\Psi_i)=UI\angle\varphi=S\angle\varphi$$
$$=UI\cos\varphi+\mathrm{j}UI\sin\varphi=P+\mathrm{j}Q$$

复功率也可表示为：

$$\overline{S}=\dot{U}\dot{I}^{*}=Z\dot{I}\cdot\dot{I}^{*}=ZI^{2}=(R+jX)I^{2}=RI^{2}+jXI^{2}$$

or $\overline{S}=\dot{U}\dot{I}^{*}=\dot{U}(\dot{U}Y)^{*}=\dot{U}\cdot\dot{U}^{*}Y^{*}=U^{2}Y^{*}$

2. 结论

(1) $\overline{S}$ 是复数，而不是相量，它不对应任意正弦量；

(2) $\overline{S}$ 把P、Q、S联系在一起，它的实部是有功功率，虚部是 无功功率，模是视在功率；

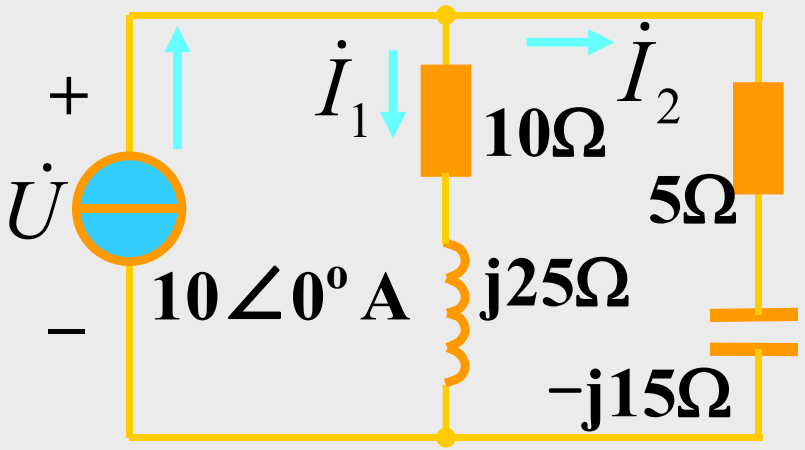
(3) 复功率满足守恒定理：在正弦稳态下，任一电路的所有支路吸收的复功率之和为零。即

$$\begin{cases} \sum_{k=1}^{b} P_k = 0 \\ \sum_{k=1}^{b} Q_k = 0 \end{cases} \longrightarrow \sum_{k=1}^{b} (P_k + \mathrm{j}Q_k) = \sum_{k=1}^{b} \overline{S}_k = 0$$

注：复功率守恒，不等于视在功率守恒 .

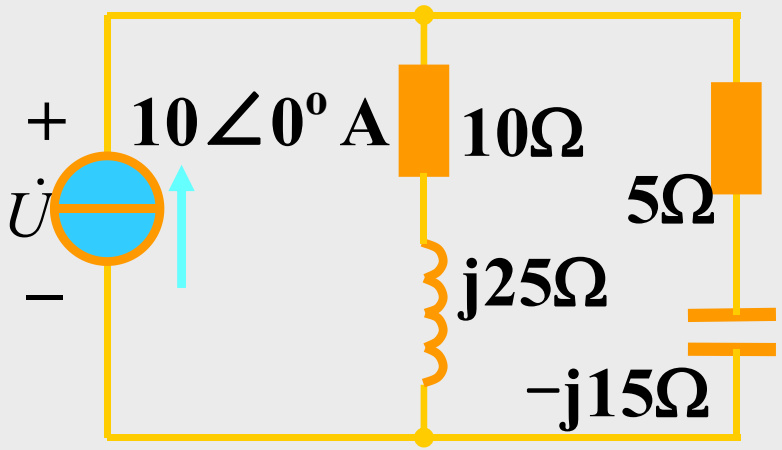
例

电路如图，求各支路的复功率。



解

$$Z = (10 + j25) // (5 - \mathrm{j}15)$$



$$\dot{U} = 10\angle 0^{o} \times Z = 236\angle(-37.1^{o}) \quad \mathrm{V}$$

$$\overline{S}_{发} = 236\angle(-37.1^{o}) \times 10\angle 0^{o} = 1881 - \mathrm{j}1425\ \mathrm{VA}$$

$$\overline{S}_{1吸} = U^{2}Y_{1}^{*} = 236^{2}\left(\frac{1}{10 + j25}\right)^{*} = 768 + \mathrm{j}1920 \quad \mathrm{VA}$$

$$\overline{S}_{2吸} = U^{2}Y_{2}^{*} = 1113 - \mathrm{j}3345 \quad \mathrm{VA}$$

$$\overline{S}_{1吸} + \overline{S}_{2吸} = \overline{S}_{发}$$

8.5 最大功率传输



$$Z_i = R_i + jX_i, \quad Z_L = R_L + jX_L$$

$$\dot{I} = \frac{\dot{U}_S}{Z_i + Z_L}, \quad I = \frac{U_S}{\sqrt{(R_i + R_L)^2 + (X_i + X_L)^2}}$$

有功功率 $$P = R_L I^2 = \frac{R_L U_S^2}{(R_i + R_L)^2 + (X_i + X_L)^2}$$

讨论正弦电流电路中负载获得最大功率P_{max}的条件。

(1) $Z_L = R_L + jX_L$可任意改变

$$P = \frac{R_L U_S^2}{(R_i + R_L)^2 + (X_i + X_L)^2}$$

(a) 先设R_L不变，X_L改变

显然，当$X_i + X_L = 0$，即$X_L = -X_i$时，P获得最大值

(b) 再讨论R_L改变时，P的最大值

当 $R_L = R_i$ 时，P获得最大值

$$P_{max} = \frac{U_S^2}{4R_i}$$

综合(a)、(b)，可得负载上获得最大功率的条件是：

$$\begin{cases} R_L = R_i \\ X_L = -X_i \end{cases} \rightarrow Z_L = Z_i^*$$

最佳匹配

(2) 若$Z_L = R_L + jX_L$只允许X_L改变

获得最大功率的条件是：$X_i + X_L = 0$，即 $X_L = -X_i$

最大功率为 $P_{max} = \dfrac{R_L U_S^2}{(R_i + R_L)^2}$

(3) 若$Z_L = R_L$为纯电阻

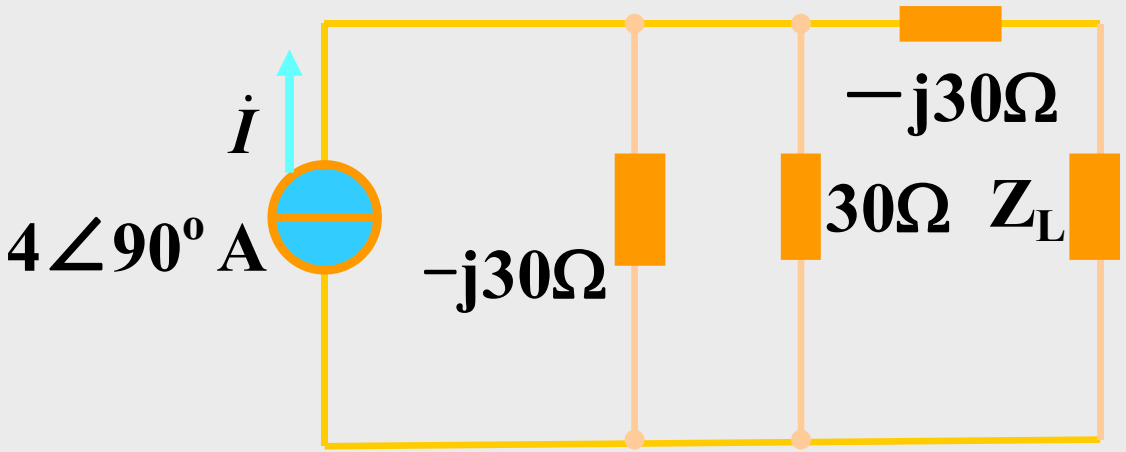
负载获得的功率为： $P = \dfrac{R_L U_S^2}{(R_i + R_L)^2 + X_i^2}$

令 $\dfrac{dP}{dR_L} = 0 \Rightarrow$ 获得最大功率条件：$R_L = \sqrt{R_i^2 + X_i^2} = |Z_i|$

模匹配

例

电路如图，求Z_L=?时能获得最大功率，并求最大功率.



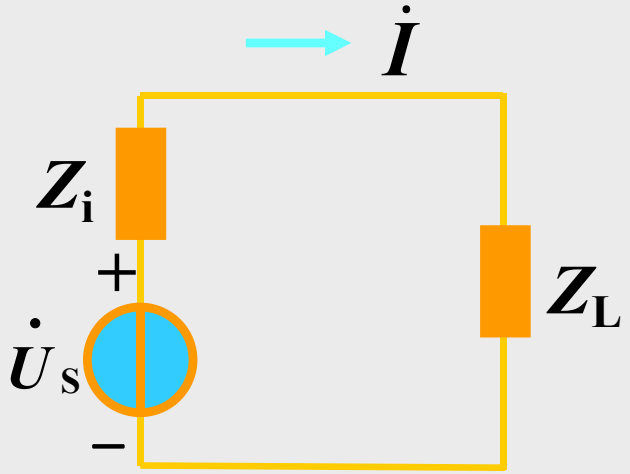
解

$$\dot{U}_S = 4j \times (-j30 \,//\, 30) = 60\sqrt{2}\angle 45^0$$

$$Z_i = -j30 + (-j30 \,//\, 30) = 15 - j45\Omega$$

当 $Z_L = Z_i^* = 15 + j45\Omega$

有 $P_{\max} = \dfrac{(60\sqrt{2})^2}{4 \times 15} = 120W$



作业

P243-P251:

9-1(a)、(b)；9-3(3)；9-5； 9-8；

9-18；9-27;

六题选做三题，本周五交

相量法、正弦稳态电路的分析
两章的作业一起交

The End